ORACLE METHOD^{SM}

PROJECT MANAGEMENT

METHOD HANDBOOK

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Project Management (PJM) Method Handbook, Release 2.6.0

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Preface

The *Project Management Method Handbook* provides an overview of the Project Management Method (PJM) in the project life-cycle. PJM is Oracle's full life-cycle approach to managing information technology (IT) projects.

The material in this book allows project managers and team members to better understand the full scope of project management effort and to plan and execute PJM tasks. Specifically, this book includes an overview of PJM, overviews and diagrams of each part of the project management life-cycle, including PJM tasks and their dependencies, and information on estimating and scheduling the tasks in PJM.

This handbook, and the Project Management Method itself, are part of Oracle Method, Oracle's integrated approach to solution delivery.

Audience

The *Project Management Method Handbook* is written for project managers and their project management team. Project managers will use this handbook on a regular basis as a reference during the project. Project management team members can use this book to gain a more thorough understanding of project management tasks and their organization at any time before or during the project.

How The Manual is Organized

This handbook consists of two parts: an overview of PJM, followed by a full treatment of the project management life-cycle.

Part I: Overview The overview presents an introduction to PJM and discusses how PJM is organized to execute a phased project. It also provides an overview of the key concepts, values, and philosophies on which PJM is based.

Part II: The Project Management Life-Cycle There are five chapters describing each category of the PJM project management life-cycle. Each category chapter consists of two main sections: an overview of the category, and a section on the approach to accomplishing that category. The overview section and the approach section provide the following information.

Overview:

- **Objectives -** describes the objectives of the category
- **Critical Success Factors -** lists the success factors of the category
- **Overview Diagram -** illustrates the processes, prerequisites, and key deliverables
- **Prerequisites -** lists task prerequisites and their sources
- **Processes -** lists and defines the processes used in that category
- **Key Deliverables -** lists and defines the deliverables for the category

Approach:

- **Tasks and Deliverables -** lists the task executed and the deliverable produced. This table also provides the responsible role and type of task. Task type definitions can be found in the glossary at the end of this book. Task types are:

  - SI - singly instantiated task (standard task)

  - MI - multiply instantiated task

  - MO - multiply occurring task

  - IT - iterated task

  - O - ongoing task

- **Task Dependencies -** illustrates the dependencies between tasks
- **Managing Risks -** provides assistance in reducing the risks associated with this category
- **Tips and Techniques -** provides guidance and helpful tips and techniques for executing each process in the category
- **Estimating -** illustrates the relative effort of tasks within the category, by role
- **Scheduling -** discusses the approach to scheduling the tasks in this category

Appendix A: Appendix A provides a description of the roles used in PJM.

Appendix B: Appendix B lists the roles used in PJM along with the tasks which call for the participation of that role.

Appendix C: Appendix C provides a listing of references and publications.

Glossary: The Glossary provides a glossary of terms used in PJM.

How to Use this Manual

The *Project Management Method Handbook* should be used as a management guidebook for managing Oracle Method based projects. This handbook should always be used in conjunction with the *Project Management Process and Task Reference*. The PJM *Process and Task Reference* provides detailed information on the tasks and deliverables that make up the PJM approach. Together these books provide a complete guide to the planning and execution of project management work.

If you are unfamiliar with PJM, version 2.5, start by reading Part I— "Overview."

Oracle Services recommends that users of all of the PJM handbooks, and the Project Management Method itself, take advantage of project management training courses provided by Oracle Education. In addition to the PJM handbooks and training, Oracle Services also provides experienced PJM consultants, automated work management tools customized for PJM, and tools for generating PJM deliverable templates.

Conventions Used in this Manual

We use several notational conventions to make this handbook easy to read.

Attention

We sometimes highlight especially important information or considerations to save you time or simplify the task at hand. We mark such information with an attention graphic, as follows:

Attention: Be sure to apply provisions for quantified risks, and contingencies for risks that are not quantifiable, to your phase workplan based on risk mitigation strategies from Control and Reporting.

For More Information

Throughout the handbook we alert you to additional information you may want to review. This may be a task section, appendix, or reference manual. We highlight these references with an easy-to-notice graphic. Here is an example:

Reference: PJM *Process and Task Reference* for detailed task and deliverable descriptions..

Web site: You can find further information on Oracle's Home web page http://www.oracle.com/

Suggestions

We provide you with suggestions throughout the handbook to help you get the most out of the method. We highlight these suggestions with an illuminated light bulb. Here is an example of a suggestion:

Suggestion: Separate the reserves in your workplan and finance plan so that transfers from the reserves to project tasks can be tracked separately.

Warning

We alert you to critical considerations or possible pitfalls to help you avoid trouble. We mark such information with a warning graphic, as follows:

Warning: If you choose to simplify or conduct the project without the initial Project Management Plan, you risk not having a point of reference for change control, and must rely heavily on verbal commitments, which can often lead to serious misunderstandings with the client and contractual disputes.

Bold Text

Bold text is designed to attract special attention to important information.

Italicized Text

Italicized text indicates the definition of a term or the title of a manual.

UPPERCASE TEXT

Uppercase text is used to call attention to command keywords, object names, filenames, and so on.

Related Publications

Books in the PJM suite include:

- PJM *Method Handbook* (this book)
- PJM *Process and Task Reference*

Your Comments are Welcome

Oracle Corporation values and appreciates your comments as an Oracle PJM practitioner and reader of the handbook. As we write, revise, and evaluate our documentation, your comments are the most valuable input we receive. If you would like to contact us regarding this or other Oracle Method manuals, please use the following address:

email: pjminfo@us.oracle.com

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PART *II* The Project Management Life-Cycle

PART

I

Overview

CHAPTER

1

Introduction to PJM

This chapter discusses the overall content and structure of Oracle's Project Management Method (PJM).

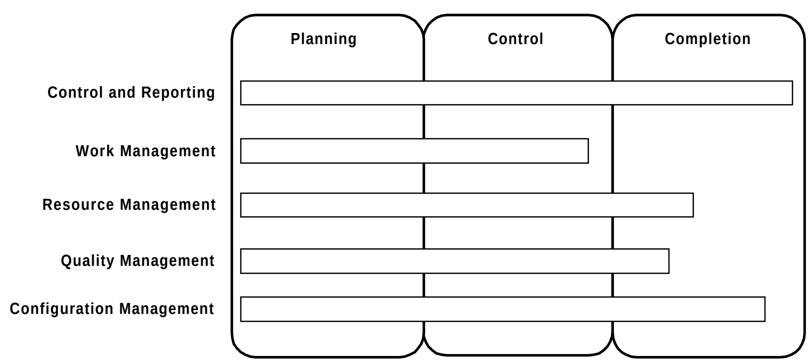

Figure 1-1 Project Management Process Overview

What is PJM?

Oracle Project Management Method (PJM) is Oracle Method's standard approach to project management for information technology projects. The goal of PJM is to provide a framework in which all types of information technology (IT) projects can be planned, estimated, controlled, and completed in a consistent manner. This consistency is necessary in an environment where projects use a variety of methods, tools, and approaches to satisfy business needs.

IT projects are characterized by a high degree of uncertainty. Information technology is a young, evolving engineering discipline which uses rapidly changing tools and techniques.

PJM addresses the unique management demands of IT projects. It focuses on the additional discipline needed to ensure that client expectations are clearly defined at the outset of the project and remain visible throughout the project life-cycle. PJM also formalizes control mechanisms to help the project team share critical project information and coordinate with external stakeholders.

PJM is designed to support a variety of types of project work. Although it has been developed for moderate to large-scale projects, PJM is also applicable to smaller efforts as well. The PJM approach can be tailored to project work performed by teams, work packages, sub-projects, pilots, and even programs.

Processes in PJM

The overall organization of PJM is expressed as a process-based methodology, which can be tailored to a project's specific needs.

A *process* is a cohesive set or thread of related tasks that meet a particular project objective. A process results in one or more key deliverables and outputs. Each process is also a discipline that involves the use of similar skills.

The five management processes are:

- Control and Reporting
- Work Management
- Resource Management
- Quality Management
- Configuration Management

These processes and their relationships are shown in Figure 1-2. Collectively, they form a complete set of all tasks required to manage an IT project. Every project involves most, if not all, of these processes, whether they are the responsibility of a consulting organization, a client organization, or a third party. The processes overlap in time with each other, and they are interrelated through common deliverables and outputs.

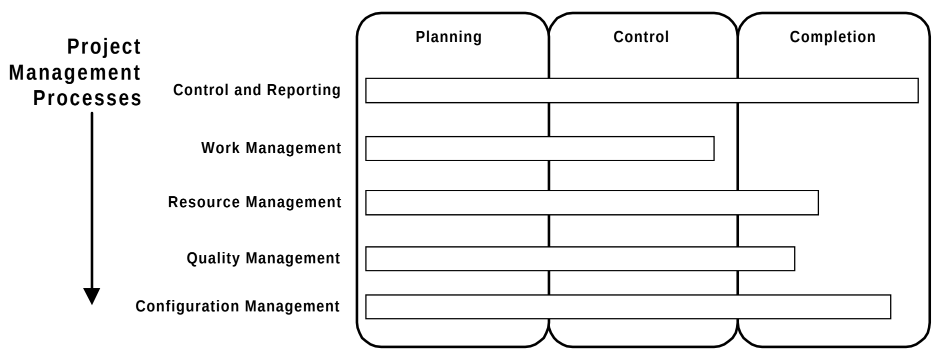

Figure 1-2 Processes in PJM

Control and Reporting

This process contains tasks that help you confirm the scope and approach of the project, manage change, and control risks. It contains guides for you to manage your project plans and report project status.

Work Management

The Work Management process contains tasks that help you define, monitor, and direct all work performed on the project. This process also helps you maintain a financial view of the project.

Resource Management

This process provides you with guidance on achieving the right level of staffing and skills on the project and on implementing an infrastructure to support the project.

Quality Management

The Quality Management process directs you to implement quality measures to ensure that the project meets the client's purpose and expectations throughout the project life-cycle.

Configuration Management

This process contains tasks that help you store, organize, track, and control all items produced by and delivered to the project. The Configuration Management process also calls for you to provide a single location from which all project deliverables are released.

Project Delivery Organization

Each PJM process organizes tasks into three groups as shown in Figure 1-3:

- Planning Tasks

  Provide the definition of the project with respect to scope, quality, time, and cost. These multiply occurring tasks also determine the appropriate level and organization of resources to execute the project.

- Control Tasks

  Performed concurrently with execution tasks. Control ensures that objectives are being met by measuring performance and taking corrective action, as needed. Ongoing Control tasks are performed continuously. Multiply instantiated Control tasks are performed as needed. Control tasks coordinate with each other by exchanging information and synchronizing their actions.

- Completion Tasks

  Completion tasks formalize acceptance of project deliverables and bring the project to an orderly end. Completion results in the satisfactory conclusion of the project.

These tasks support the project’s *execution* tasks, taken from the remaining project processes responsible for producing the business deliverable products and services of the project. These task groups are organized into a general life-cycle model known as the *project delivery model*. This model can be used to organize virtually any kind of project work.

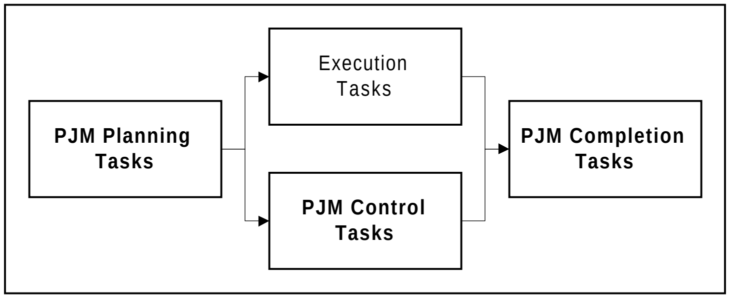

Figure 1-3 Project Delivery Model

Project Management Life-Cycle

Dividing a project into phases provides a higher degree of management control and reduces uncertainty. The end of each phase reflects the completion of a major set of project deliverables which can be reviewed and signed-off by the client. Each phase break also represents a strategic point in the project, providing an opportunity to confirm the client's business needs.

Collectively, a project's phases are known as the *project life-cycle* shown in Figure 1-4. The PJM tasks are organized into five phases:

- **Project Planning**

  Tasks in this category encompass the definition of the project with respect to scope, quality, time, and cost. Project planning tasks also determine the appropriate organization of resources and responsibilities to conduct the project.

- **Phase Planning**

  This category consists of tasks which update project plans and procedures for the phase.

- **Phase Control**

  Tasks in this category execute concurrently with phase product delivery, and perform project monitoring, directing, and reporting functions during the phase.

- **Phase Completion**

  These tasks conclude and secure sign-off of a phase.

- **Project Completion**

  Tasks in this category result in the satisfactory conclusion of the project and settlement of all outstanding issues prior to shutting down the project.

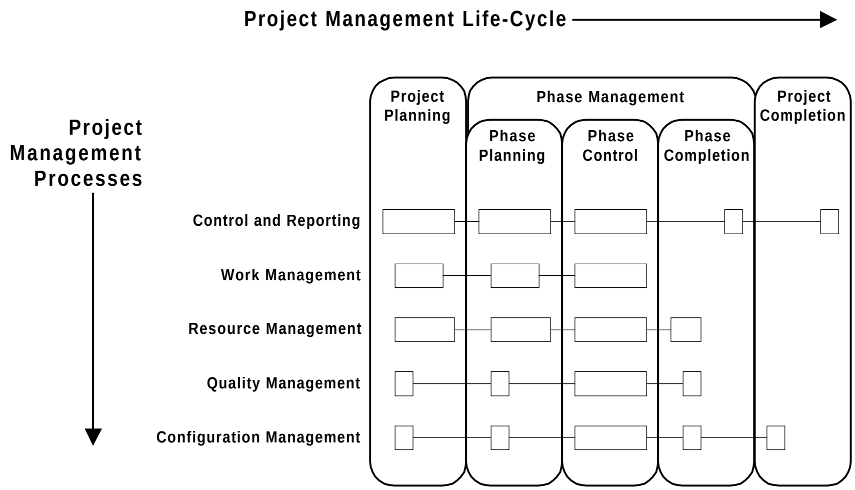

Figure 1-4 Project Management Processes and Life-Cycle

Project Life-Cycle Integration

Figure 1-5 shows how PJM tasks are integrated with a delivery method in a hypothetical model of a project workplan. Project Planning and Completion are each performed once, at the beginning and end of a project, respectively. Phase Management (Planning, Control, and Completion) is performed for each phase of the project.

PJM defines dependencies such that Phase Management tasks do not fall on the project's critical path, except at the beginning and end of the project. Project delivery tasks, represented by bars labeled Execution, form the project's critical path along with PJM Project Planning and Project Completion.

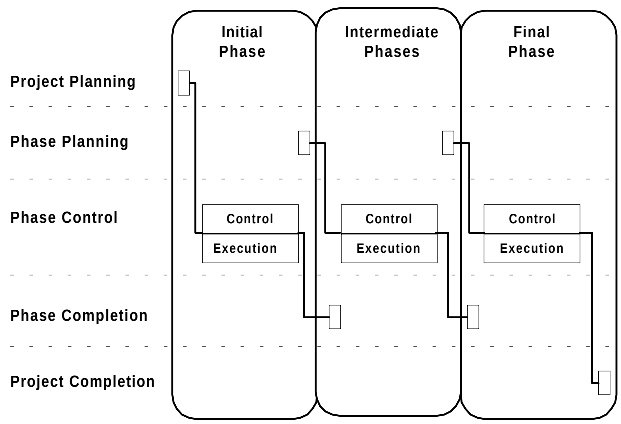

Figure 1-5 Organizing PJM in the Project Life-Cycle

Estimating and Organizing Project Management

Estimating and organizing project management involves three factors:

- Project Management Effort
- Consulting-Client Relationship
- Project Management Staffing

Project Management Effort

Oracle's project experience indicates that total project management effort typically ranges from 10% of project effort for small projects to as much as 25% for the largest projects. Project management relative effort increases as the project size increases because of increasing phase control complexity and coordination among full time project management team members. Project duration also affects project management effort relative to total project effort, since Phase Management occurs during the entire phase.

The table below summarizes Oracle's experience for managing medium sized projects. Each cell represents the percent of project management effort of a PJM process in a life-cycle category. This table shows Phase Control as the most consuming category of project management, with Work Management the most consuming process.

PROCESS (% of PJM effort)	Project Planning	Phase Planning	Phase Control	Phase Completion	Project Completion	TOTAL
CR - Control and Reporting	2%	3%	24%	1%	0%	**31%**
WM - Work Management	2%	8%	23%	0%	0%	**33%**
RM - Resource Management	2%	2%	5%	2%	1%	**12%**
QM - Quality Management	0%	0%	6%	2%	0%	**9%**
CM - Configuration Management	0%	0%	13%	2%	1%	**16%**
TOTAL	**6%**	**14%**	**71%**	**7%**	**2%**	**100%**

You can estimate PJM work effort using either top-down or bottom-up estimating techniques. The most reliable approach to estimating PJM work effort is to calculate a bottom-up estimate. A bottom-up estimate can only be developed from a work breakdown structure of PJM tasks that contains estimating factors. The factors are used in an estimating formula that produces a task estimate for each task. In PJM, project complexity and total project execution effort are the key estimating factors used in a bottom-up estimate.

These same factors can also be used to calculate a top-down PJM estimate. For example, a function point analysis gives an estimate of execution effort. You can then use the values in the table above as a guide for allocating PJM effort, once you have arrived at a suitable project management-to-execution effort ratio for your project.

☞ **Attention:** The percentages for Phase Planning, Phase Control, and Phase Completion represent the *total* amount of effort across *all* project phases for the entire project. The amount of effort you allocate from these categories for each phase varies with the total effort and duration of the phase.

Consulting-Client Relationship

The people who have influence over the products and conduct of the project may be drawn from within the organization, supplied by an outside organization, or a combination of both. A contract may or may not be involved. In PJM, the *consultant* and the *client* represent the two parties which together form the project management team responsible for the project's success.

The client represents the customer organization, or primary beneficiary of the project's deliverables, as well as the acquirer, or funding source for the project. The client is also assumed to be capable of providing both physical and human resources for the project.

The consultant represents an information services provider organization with management structure and systems. This organization may be either a profit center, performing the project for a profit, or a cost center, sharing project costs with the client. It is made up of practices, or business units, which supply consulting staff resources and sub-contractors to the project.

Note that the tasks performed in reaching and maintaining a contractual agreement between the client and consultant are not covered in PJM.

PJM assumes that a contract may be established prior to the start of the project, and identifies where contractual impacts can occur during the project. A contract is not a prerequisite for the use of PJM.

PJM also assumes that both the client and the consultant have internal management policies governing project conduct. Tailor these aspects of the client-consultant relationship to your project's specific situation.

Key Project Management Roles

The key management roles performed by the client in PJM are *the project sponsor* and *client project manager*. The project sponsor is the client role that holds the budget for the project, and may be an individual or a committee. The project sponsor ensures organizational commitment to the project and validates project objectives. The client project manager is expected to be assigned to the project where client commitments or business interests require a daily client management presence. This role is responsible for providing client resources, resolving problems, and monitoring the consultant's progress.

The key management roles performed by the consultant in PJM are the consulting business manager and the project manager. The consulting business manager role represents the consulting manager whose practice is responsible for the successful execution of the project. The consulting business manager also represents the consultant if a contractual agreement exists with the client. The project manager is the consultant role which is held ultimately responsible for the project's success or failure. The project manager must manage the various aspects of time, cost, scope, and quality to satisfy client expectations and meet the business objectives of the consulting practice, while providing challenging opportunities to project staff.

Project Management Staffing

This diagram illustrates a typical project management organization. The roles depicted in the organization chart are those that are assigned responsibilities to perform PJM tasks. Refer to Appendix A, PJM Roles, for a complete definition of each role.

Staffing involves two factors:

- Role Allocations to Staff
- Multi-Site Project Considerations

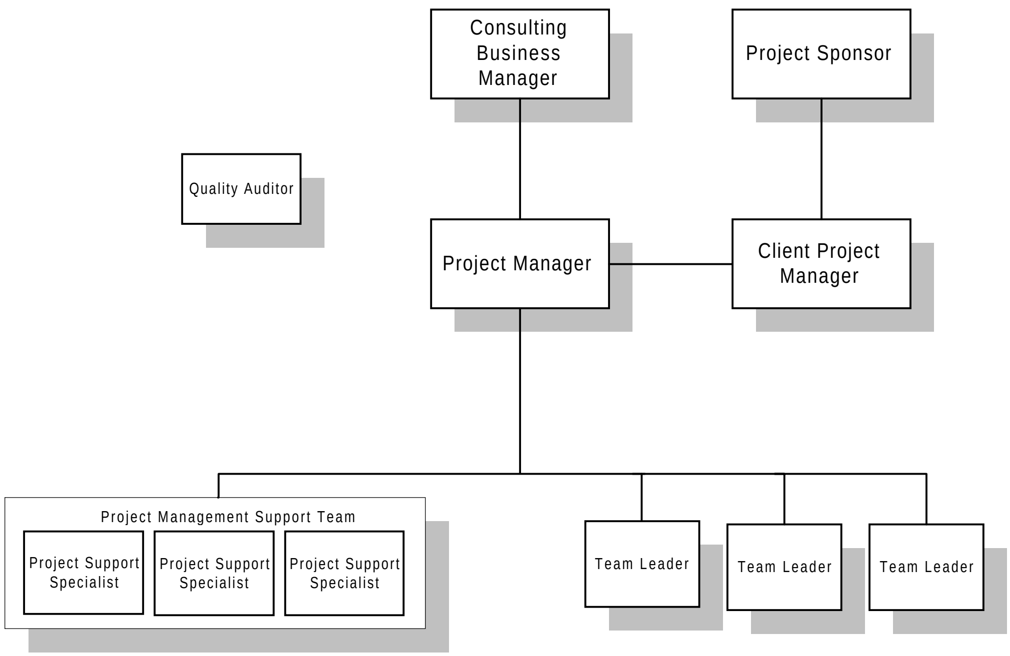

Figure 1-6 Staffing for Project Management

Role Allocations to Staff

Each role defined in the project management support team will only be assigned to different people on medium-sized projects or larger. On the largest projects, there may even be more than one person performing each of these roles, and the team will be organized into a *project office*, with a manager.

On smaller projects, the project manager will assume most of the responsibilities of the project support team. The first responsibility the project manager should relinquish as project size increases is that of configuration manager. This role is frequently assigned to a senior person performing a technical support role, such as a system administrator.

The responsibility for quality management is only a full-time position on large-scale projects. The Quality Auditor should not report to any of the project team due to a potential conflict of interest. The quality auditor is a role independent of the project as shown on the staffing chart.

There are other organizations that are commonly employed on larger projects to facilitate management communication and decision-making:

- Steering Committee

  This organization is usually chaired by the project sponsor or senior client decision-maker. Its purpose is to provide the project with strategic direction, resolve change requests and issues affecting scope, approve contract changes, and direct coordinating client actions outside of the project. The consulting business manager is also normally a member of the steering committee.

- Change Control Board (CCB)

  The CCB is an internal project organization the purpose of which is to review and resolve change requests. The CCB is chaired by the project manager and includes the client project manager, project administrator, configuration manager, and team leaders. The CCB normally escalates changes affecting scope to the steering committee.

- Issue Review Board (IRB)

  The IRB is organized to resolve issues and manage risks where a regular, dedicated meeting is deemed necessary. It is staffed similarly to the CCB, and can be combined with it.

Multi-Site Project Considerations

Multiple site projects require a higher level of project administration and control to coordinate the tasks and to leverage common deliverables between projects. In a multiple site project, you will need to position site coordinators as part of your project management team. These people also ensure that there is consistency in the delivery and presentations of work, use of techniques and approach, use of standards and guidelines, and interpretation of enterprise wide strategies.

Another important role that coordinators perform is facilitating the technical strategies between related sites. This role calls for a more formal exchange of technical information and status review. These site coordinators will also distribute software and documentation to multiple data centers.

CHAPTER

2

Key PJM Concepts

This chapter describes key concepts and philosophies on which Oracle's Project Management Method (PJM) is based. The topics discussed are:

- Fundamental Values of Project Management
- The Project Manager
- The Golden Rules of Project Management

Fundamental Values of Project Management

Information technology (IT) projects remain inherently risky, because technology and business needs continue to change. Oracle emphasizes clear and practical methods and an environment of partnership between clients and suppliers to create a foundation for success.

Oracle Corporation needs project teams who have experienced successes and setbacks to build upon this foundation. The project teams must take an assertive approach to managing risks while striving to achieve the business benefits for the client.

Figure 2-1 depicts the fundamental values of Oracle's approach to managing projects.

Figure 2-1 PJM Fundamental Values

The Project Manager

In PJM, the *Project Manager* is a consulting role critical to the success of the project. The project manager role is also associated with the person who is ultimately held accountable for the success or failure of the project. In this section we will examine the responsibilities, skills, and authority you should consider if you are assigned as a project manager.

Since every project is different, clarify your specific responsibilities and authority at the start of the project so there is no confusion about the services that you are expected to provide. In addition, decide what responsibilities and authority you will delegate to others in your project management team, such as team leaders and project coordinators.

Project Manager Responsibilities

As the project manager, you represent the project to both the client and consulting management, and expect to be held responsible for satisfying both of their expectations regarding the project. To do this, you need to understand the project's business objectives from both viewpoints. You are also responsible to the resource providers in your consulting organization for using their resources effectively and to the staff working on the project to provide them with challenge and personal growth opportunities.

You will undoubtedly be faced with conflict between the different objectives of these parties as well as various other stakeholders to the project. A key part of your responsibilities is to face outwards from the project, handling political conflicts and issues and ensuring they do not impede project progress.

As the project manager you are responsible for planning the project, resourcing that plan, and monitoring and reporting the project's progress according to the plan. Obtain any physical resources required for the project, recruit staff, and, if necessary, dismiss them. You are responsible for ensuring the quality of your project's deliverables, and that quality actions are performed in accordance with the project's Project Management Plan.

You agree on the scope of the project with the client, and ensure that the project remains within the agreed scope. You also define or approve all of the strategies your project management team will use to accomplish the project's stated objectives. You should expect to review all key deliverables, particularly those from the initial phases of the project.

Characteristics of the Project Manager

As the project manager, you face two central challenges:

- Deciding what to do, despite uncertainty, risk, and an enormous amount of potentially relevant information.
- Getting things done, through a large and diverse set of people, despite having limited direct control over them.

The key characteristics required of you in meeting these challenges are summarized below.

Personal Characteristics

The challenge of deciding what to do requires the application of personal characteristics:

- show common sense
- be organized
- focus on the future
- use judgment
- maintain perspective of the key objectives

Interpersonal Characteristics

The challenge of motivating others to get things done requires the application of interpersonal characteristics:

- Lead the team

  Be an integrator, bringing in a variety of people from many areas into a cohesive multi-disciplined team.

- Assert power

  Build confidence and gain respect, but be expedient in the use of authority when the situation demands.

- Show drive, stamina, and stability under pressure.
- Match management style to situation

  Use appropriate negotiation and conflict-handling skills.

- Communicate effectively
- Be sensitive to culture and politics

Project Management Skills

General management skills provide much of the foundation for building project management skills and are essential to project management. These skills are:

- Leading
- Communicating
- Negotiating
- Problem Solving
- Managing Conflict
- Influencing the Organization

Leading

While managing is primarily concerned with consistently producing results, leading focuses on establishing direction, aligning people, and motivating and inspiring. Leadership should not be limited to the project manager. It should be demonstrated at all levels of the project management team.

Communicating

Communicating involves the exchange of information, and it can take many different forms such as written, oral, internal, external, formal, informal, vertical, horizontal. Effectively communicating requires the application of techniques such as:

- use of sender-receiver models
- choice of media
- writing style
- presentation
- meeting management

Negotiating

Negotiating involves conferring with others in order to come to terms or reach an agreement. Negotiations occur within many of the management tasks in PJM. During the project, you can expect to negotiate for any or all of the following:

- scope, cost, and schedule objectives
- changes to scope, cost, or schedule
- contract terms and conditions
- assignments
- resources

Problem Solving

Problem solving is the result of problem definition and decision-making. Problem definition requires distinguishing between causes and symptoms. Decision-making includes analyzing the problem to identify viable solutions, making a choice from among them, and then implementing that choice. Decisions also involve a time element: the right decision is the best solution at the time it must be made.

Managing Conflict

Managing conflict involves arbitrating among differing outlooks, priorities, attitudes, viewpoints, and orientations, to arrive at the best solution for the project. Avoiding conflict during a project will ultimately lead to a loss of control over it. Know how and when to employ various conflict management techniques, such as:

- Smoothing: you put things in their proper proportions and emphasize understanding and commonality of objectives and viewpoints.
- Compromising: you find options or solutions that are acceptable to both parties and get them to agree on the compromise.
- Forcing: you use authority to force an option or situation on the conflict.
- Withdrawing: if resolving the conflict may be more damaging to the project than the conflict itself.
- Confronting: you intentionally create a situation to bring out a conflict that was below the surface.

Influencing the Organization

Influencing the organization involves the ability to get things done. It requires an understanding of the formal and informal structures within the project organizations involved. Both power and politics are used in a positive sense, to get people to do things they would not otherwise do, within a group that may have widely divergent interests.

Suggestion: Oracle Services offers two internal workshops that teach the application of general management skills in a project setting: *Introduction to Project Management in Oracle* and *Introduction to Team Leading in Oracle.*

Project Stakeholders

There are three main groups of stakeholders whose interests you must balance to ensure project success:

- The Client
- The Consulting Practice
- The Project Staff

The Client

The client wants a system that fulfills a business need(s) and that is acceptable to the client's staff, customers, and suppliers. The client wants to feel that value has been received for the money paid and that your organization is a supplier with which business should be conducted again.

The Consulting Practice

Your practice wants a successful project that enhances the practice's reputation. It wants a profitable project with a reasonable return on investment. It also wants a client that is willing to act as a reference site and that will offer repeat business.

The Project Staff

The staff on a project want to develop personally. Each project should give them opportunities to enhance their standing within your practice and the industry. They wish to learn and profit by the experience, and they also want to enjoy their work.

Project Manager Interactions

Accurate, timely communication from you to your stakeholders is an important factor in the ultimate success of your project. The following are key interactions with other PJM roles:

- Interaction with the Project Sponsor
- Interaction with the Client Project Manager
- Interaction with the Project Staff
- Interaction with the Consulting Business Manager

Interaction with the Project Sponsor

- Inform senior management of critical issues.
- Recommend changes in organization or project scope where necessary for the success of the project.
- Request additional funding and provide justification.
- Identify obstacles in meeting critical success factors.

Suggestion: Do not overload senior management with details. Time is always limited and you need to focus only on important points. Prepare a plan or options for each issue you raise. Offer measurements and time frames for meeting objectives.

Interaction with the Client Project Manager

- Communicate project status, including consulting budget-to-actual status.
- Document and update financial status and forecasts based on actuals and estimates-to-complete.
- Address business, system, and staffing issues.
- Address scope issues in a timely way.

Suggestion: Be sensitive to how you communicate with the client. Help the client to succeed, rather than finding fault. Always define what needs to be done, and who should be responsible. Status reports and meetings are appropriate methods of communication. Leave time to research issues and verify facts. Be careful not to omit appropriate people from your distribution list.

Interaction with the Project Staff

You also provide detailed information to the project staff at periodic progress reviews and during informal discussions. During these sessions you:

- Summarize accomplishments, plans, and issues for detailed tasks.
- Identify near-term goals and required deliverables.
- Resolve outstanding issues and problems.
- Get to know your project members and look for hidden problems.

Suggestion: Clearly set expectations and provide specific direction where necessary. Be explicit about who is responsible for a particular task or action item. During meetings defer long, detailed discussions to a later time in order to cover all areas on the agenda. Ensure that meeting minutes clearly show to all who is accountable for the completion of assigned tasks.

Interaction with the Consulting Business Manager

- Submit resource requests and negotiate for staff.
- Understand the contractual agreement.
- Obtain approval for contract change proposals.
- Plan and review project work and financial status.
- Ensure that deliverables are archived according to policy.
- Provide feedback on successful projects, such as client environment, benefits, and contacts.
- Identify new consulting opportunities with the client.

The Golden Rules of Project Management

Projects, by definition, are unique. However, there are certain principles that are common to the management of successful projects, known as the *Golden Rules of Project Management*:

- Start Right
- Know Your Client
- Define the Project Scope
- Plan to Reduce Uncertainty
- Manage the Risks
- Field a Winning Team
- Maintain Team Commitment
- Communicate with Honesty and Conviction
- Use the Project Management Plan
- Produce Formal Documentation
- Plan for Completion

Keep these important principles in mind when you refer to the specific tasks, deliverables, and responsibilities described in PJM, or when you use Oracle Method on a daily basis as a member of a project management team.

Start Right

If you start badly, it is always difficult and often impossible to recover.

Projects are no exception. Planning is the most critical part of a project, establishing definition and commitment.

Projects that fail usually do so because certain fundamental principles are not followed from the moment the project is started. It is particularly important during the start of a project that these rules should be observed and the organization and framework established to ensure that the discipline is maintained throughout the life of the project.

Know Your Client

Learn who has the most to gain from the success of the project and, conversely, who has the most to lose from failure.

Get to know the project stakeholders and, in particular, who is funding the project. These two parties usually are not the same people. Understand their spheres of influence, agree on when and how their support will be called upon, and what the project will expect of them.

A project manager will fail without being given consistent objectives, clear guidance, and strong direction from the executive management of all parties involved.

One of the most effective ways of achieving all party executive management support and encouraging open communication is through a steering group or project board comprising the stakeholders and main beneficiaries.

Remember that *people* buy, build, and use systems. If you support their objectives, their positions, and their needs, you will develop real business benefits for your client and will achieve the satisfaction of a job well done.

Define the Project Scope

Reach agreement with the client on a precise definition of the scope of work and terms of engagement.

It should be sufficiently detailed to enable you to derive an achievable plan and basis for change control. You should understand how consulting's scope of work relates to the client's business objectives and the key benefits the client expects to achieve from it.

Plan to Reduce Uncertainty

Do not try to plan what you do not know.

The simplest and probably the most effective way of managing risk is to plan short, detailed cycles. Plan in detail the current phase, then plan in general all future phases, including the key milestones. State all the assumptions made in preparing the plan, especially those relating to the later phases, and get them validated by the client.

Control the future, but remember, only the immediate future can be predicted with a reasonable degree of certainty.

Manage the Risks

Determine the key risks, analyze their impact, define containment strategies, and establish contingency plans.

Monitor risks at every stage, and be able to recognize at the earliest possible time when to invoke a contingency plan. Every change in requirements, scope, approach, or design is a risk that needs to be assessed and incorporated into the plan

Field a Winning Team

Staff your team with an appropriate blend of individuals.

You will need to apply a considerable degree of assertion in order to obtain the best possible people for the job at the right time, whether they are internally or externally sourced. People should always be selected on their merits first, their ability to adapt rapidly second, and on their personality fit with the team third.

Maintain Team Commitment

Spend time together and encourage a sense of belonging and ownership of the project goals.

Explain the project goals and how they relate to the client's objectives. Describe how they will be achieved by the team. This is especially important for those who join in the middle of the project to help them into the team.

Gain and maintain commitment from the whole team by helping each of them to understand and achieve in their role. Encourage individual accountability, but let team members know what your expectations are and agree on how their performance will be measured.

Build the best possible work environment, always be there for the team, and make the assignment an enjoyable experience for everyone. Take the time to thank individuals, especially for effort beyond the call of duty. Appreciation is the leading motivator.

Communicate with Honesty and Conviction

Set standards with the stakeholders and the team on communications and progress reporting, adhere to them, be honest, and do it frequently.

Set for everyone involved the expectation of "*no surprises*". This requires you to be on top of the job at all times. It is not sufficient to rely on formal reports; ad hoc status checks, presentations, or briefs will be called for at short notice, so be prepared.

Adopt a conflict minimization strategy but do not expect to avoid conflict completely. Be prepared for it to happen, resolve it, and try to reach a win-win solution.

Use the Project Management Plan

The Project Management Plan is the medium for communicating exactly how consulting intend to fulfill our obligations on the project.

Use established methods and tools. Oracle Method provides a complete life-cycle methodology for the implementation of business goals through IT. It is especially important that your practice recommendations are followed for estimating, planning, tracking, quality assurance (QA), change control, issue management, and documentation.

The Project Management Plan should state what policies, procedures, and standards apply to the project and is the baseline against which quality audits will be performed. It is therefore essential that the client's approval be obtained for this document.

Produce Formal Documentation

Any commitment by you or the client should be agreed to in writing.

Document all deliveries, agreements, decisions, issues, resolutions, actions, and file them with all correspondence and minutes of meetings. Written evidence can be used to avoid disputes as well as settle them. A good and fair contract provides mutual protection to both consulting and client.

Plan for Completion

When you walk away after all the tests have been signed off, remember: the system you have gone through hell and high water to give birth to has just started the rest of its life.

The effective and efficient closure of a project can make a significant contribution to its success. Project Completion is the final phase of the change process brought about by the project. Plan early for it. Final impressions are the ones that you leave with people.

Prepare a project end report. Review what went right, what went wrong, and what could have been done better. This self appraisal will be invaluable for those who undertake future projects and is a key method by which others will learn to improve.

PART

II

The Project Management Life-Cycle

CHAPTER

3

Project Planning

This section describes the Project Planning life-cycle category of PJM. The goal of Project Planning is to define the project objectives and approach with respect to scope, quality, time, and cost.

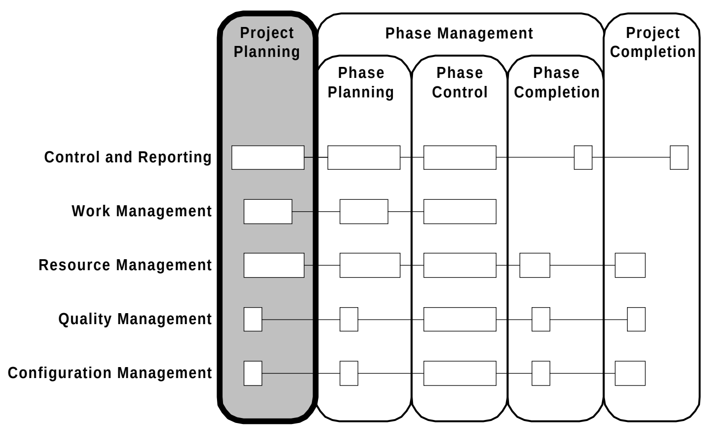

Figure 3-1 Context of PJM Project Planning

Overview

This section provides an overview of Project Planning. Specific topics discussed are:

- Objectives
- Critical Success Factors
- Overview Diagram
- Prerequisites
- Processes
- Key Deliverables

Objectives

The objectives of Project Planning are to:

- Establish the project scope, technical and business objectives, and resources and schedule required to accomplish the project objectives.
- Develop a baseline Workplan to educate the client and determine project resource requirements.
- Prepare a financial profile of the project which will be used to monitor and control finance performance.
- Obtain client and consulting management approval to proceed with execution of the project.
- Determine the measures which will be used on the project to measure and maintain the quality of project processes and deliverables.

Critical Success Factors

The critical success factors of Project Planning are:

- Scope, objectives, and approach are agreed on and understood by all parties.
- Project culture and climate are established conducive to a win-win philosophy.
- Risks are identified and containment measures are put in place.
- The client accepts the project Workplan in the context of the project's scope and risk assessment.
- The client understands the obligation to provide resources to support the project Workplan.

Overview Diagram

This diagram illustrates the prerequisites, processes, and key deliverables for Project Planning.

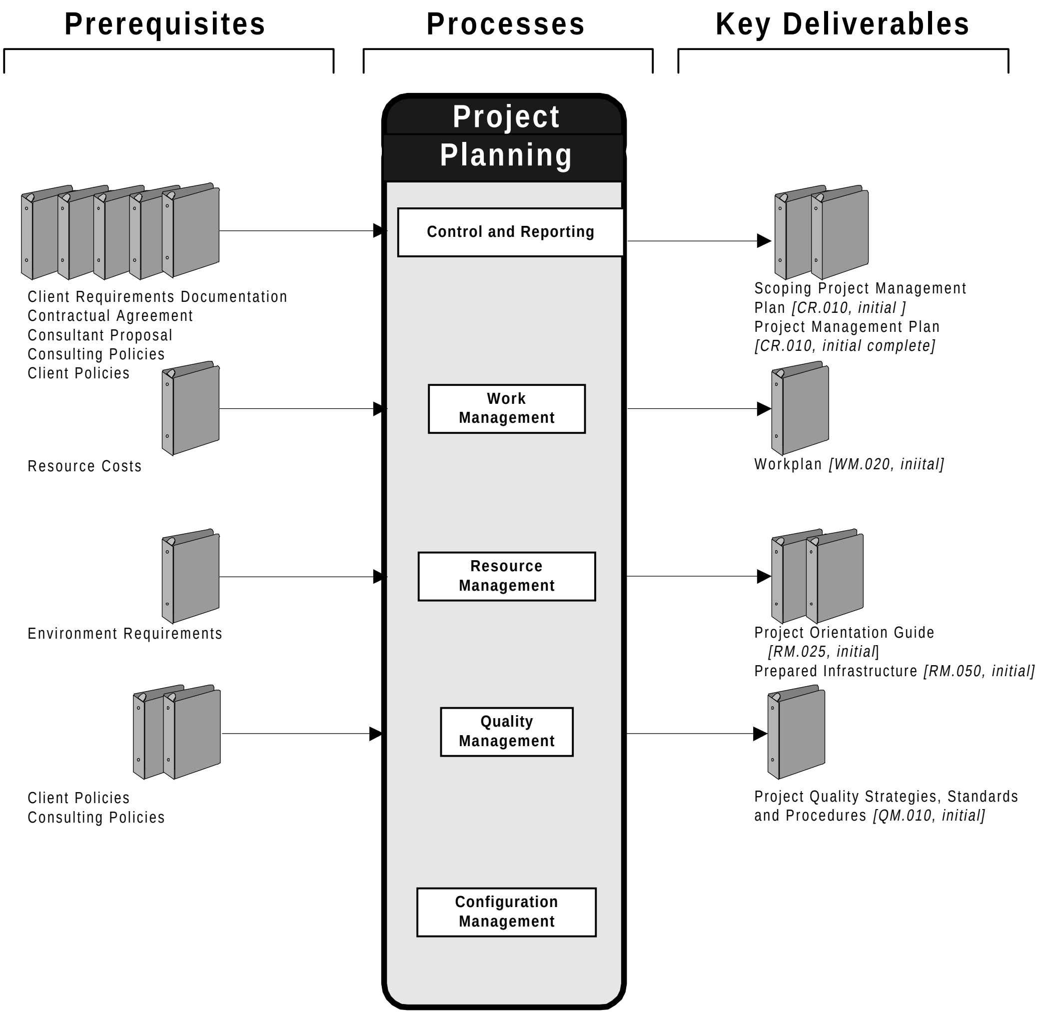

Figure 3-2 Project Planning Overview

Prerequisites

Use these prerequisites if they exist prior to beginning the project. Otherwise, you will need to determine whether their absence either prevents you from completing Project Planning tasks or creates any project risks.

Prerequisite	Source
Client Requirements Documentation	Client
Consultant Proposal	Bid Manager
Contractual Agreement	Bid Manager
Client Policies	Client
Consulting Policies	Consulting Business Manager
Resource Costs	Consulting Business Manager
Environment Requirements	Execution Processes

Processes

The processes used in this phase are:

Process	Description
Control and Reporting	Secure consulting and client agreement on project scope, objectives, and approach, and publish the Project Management Plan.
Work Management	Estimate project effort, construct project Workplan, and Finance Plan.

Process	Description
Resource Management	Plan and establish core project staff. Plan and establish core project infrastructure.
Quality Management	Document quality arrangements for the project.
Configuration Management	Plan overall approach to controlling and releasing documents and deliverables.

Key Deliverables

The key deliverables of Project Planning are:

Deliverable	Description
Project Management Plan *[CR.010, initial scoping]*	Background, objectives, scope, constraints, and assumptions for the project in terms of the client's organization and Oracle's involvement.
Project Management Plan *[CR.010, initial complete]*	The compiled set of management approaches, standards, and procedures by which the project will be managed.
Workplan *[WM.020, initial]*	A high-level network of interdependent tasks representing all project work, with staff work assignments and schedules consistent with the Project Management Plan.
Project Orientation Guide *[RM.025, initial]*	Contains all of the policies and procedures for the engagement.

Deliverable	Description
Prepared Infrastructure *[RM.050, initial]*	The set of physical resources such as office facilities, computer hardware, and software, providing the backbone for all project environments needed to execute the project.
Quality Management Strategies, Standards, and Procedures *[QM.010, initial]*	Specific standards and procedures needed to amplify high-level statement in the Project Management Plan, such as a specific procedure for auditing.

Approach

This section describes the approach for Project Planning. Specific topics discussed are:

- Tasks and Deliverables
- Task Dependencies
- Managing Risks
- Tips and Techniques
- Estimating
- Scheduling

Tasks and Deliverables

This table lists the tasks executed and the deliverables produced during Project Planning. Iterated tasks are executed during Project Planniing for the initial phase and during each subsequent phase of the project. Processes are indicated by shaded bars.

ID	Tailored Task	Deliverable	Type
Control and Reporting			
CR.010	Establish Scope, Objectives, and Approach	Scoping Project Management Plan *[CR.010, initial]*	IT
CR.020	Define Control and Reporting Strategies, Standards, and Procedures	Control and Reporting Strategies, Standards, and Procedures *[CR.020, initial]*	IT
CR.030	Establish Management Plans	Project Management Plan *[CR.010, initial complete]*	IT
Work Management			
WM.010	Define Work Management Strategies, Standards, and Procedures	Work Management Strategies, Standards, and Procedures *[WM.010, initial]*	IT
WM.020	Establish Workplan	Workplan *[WM.020, initial]*	IT
WM.030	Establish Finance Plan	Finance Plan *[WM.030, initial]*	IT

ID	Tailored Task	Deliverable	Type
Resource Management			
RM.010	Define Resource Management Strategies, Standards, and Procedures	Resource Management Strategies, Standards, and Procedures *[RM.010, initial]*	IT
RM.020	Establish Staffing and Organization Plan	Staffing and Organization Plan *[RM.020, initial]*	IT
RM.025	Create Project Orientation Guide	Project Orientation Guide *[RM.025, initial]*	IT
RM.030	Implement Organization	Prepared Organization *[RM.030, initial]*	IT
RM.040	Establish Physical Resource Plan	Physical Resource Plan *[RM.040, initial]*	IT
RM.050	Establish Infrastructure	Prepared Infrastructure *[RM.050, initial]*	IT
Quality Management			
QM.010	Define Quality Management Strategies, Standards, and Procedures	Quality Management Strategies, Standards, and Procedures *[QM.010, initial]*	IT
Configuration Management			
CM.010	Define Configuration Management Strategies, Standards, and Procedures	Configuration Management Strategies, Standards, and Procedures *[CM.010, initial]*	IT

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Task Dependencies

This diagram shows dependencies between tasks in Project Planning.

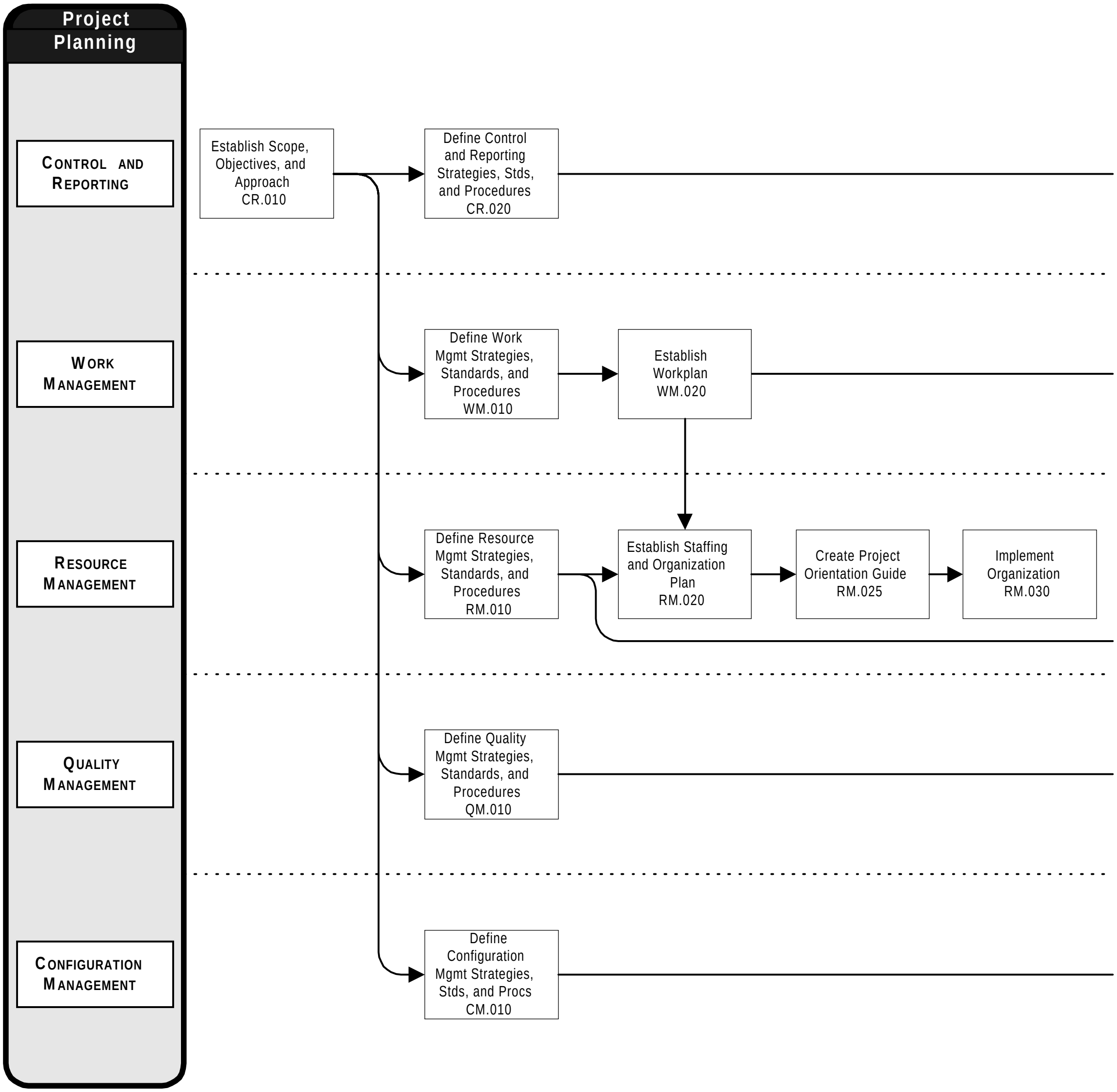

Figure 3-3 Project Planning Task Dependencies

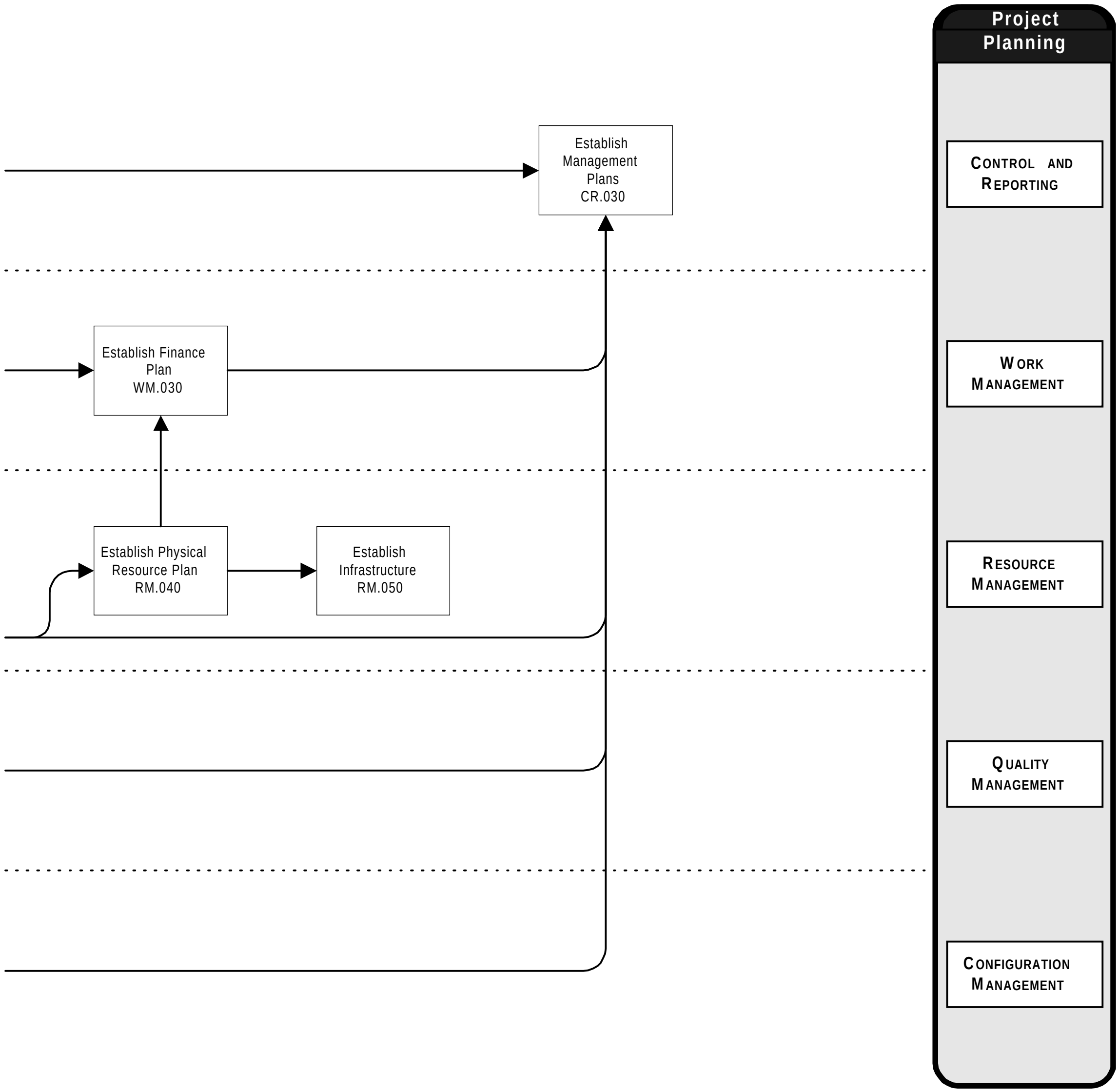

Figure 3-3 Project Planning Task Dependencies (cont.)

Managing Risks

The most likely areas of risk during Project Planning are the following:

- The definition of project scope is unclear or imprecise.
- No project sponsor exists.
- Risk contingencies and mitigation strategies are not established or not reflected in project plans.
- Key project stakeholders and their expectations are not adequately considered.
- Responsibilities for providing project resources are not clearly established.

Try to mitigate these risks in the following ways:

- Use the initial scoping Project Management Plan to clearly define the project's deliverables and underlying assumptions. Use the document also as the means for resolving differences among project stakeholders.
- Ensure that the client's organization and objectives for the project are well understood, and client stakeholders are identified. Include the project sponsor and client decision makers on the project organization chart and agree on how risks, issues, and scope changes will be jointly managed during the project. Document the information in the Project Management Plan.
- Perform a thorough review of risks identified during the bid. Reserve realistic provisions in the project Workplan and finance plan for each quantifiable risk and adequate contingencies for risks that are not quantifiable.
- Include all anticipated requirements for physical resources in the Physical Resource Plan and get the client's agreement for those parts of the project infrastructure for which the client will be responsible.

Tips and Techniques

This section includes advice and comments on each process.

Control and Reporting

Use Control and Reporting tasks to understand and influence the expectations of key project stakeholders. Use the Control and Reporting deliverables to document and communicate your vision of the project to the client, consulting management, and the project team. Evaluate the proposal as your starting point for Project Planning. How have expectations been set? Have there been changes since the proposal was accepted?

Set Expectations Early

The initial scoping Project Management Plan covers all areas of known risk, working policies, agreed approaches, communication, and implementation strategies. Consulting proposals should already include some of these topics. However, in creating the original proposal, assumptions may have been used. In creating the initial scoping Project Management Plan during Project Planning, you must verify all the documented and undocumented assumptions. The contents of the initial scoping Project Management Plan should be based on current information and *facts*.

⚠ **Warning:** You must prepare the initial scoping Project Management Plan. To conduct the project without the initial Project Management Plan, you risk not having a point of reference for change control, and must rely heavily on verbal commitments, which can often lead to serious misunderstandings with the client and contractual disputes.

In some cases, clients may need clarification and justification before approving deliverables. The initial scoping Project Management Plan can be used to explain why certain methods, approaches, and techniques were used. If organized correctly it can serve as the vehicle for justifying why and how project tasks are being performed. Within this context, the planning for deliverable reuse should be established to increase productivity and reduce costs as well as risks. Deliverable reuse planning is an integral component of the initial scoping Project Management Plan since it will assist in the establishment of client expectations. Reuse of deliverables will help to reduce the risk of the project since these provide actual examples of previous project work.

Obtain Initial Approvals

Arrange for a review from a consulting support organization, if possible upon starting the project. Review scheduling and the review itself are components of QM.045.. The purpose of the review is to determine the overall health of a project and its prospects for success, check the completeness of plans from a risk viewpoint, and ensure that a clear understanding of project control and completion procedures exists with the client.

Suggestion: Use startup meetings to communicate the plans to the project stakeholders. Consider separate meetings for client management and project staff. A formal presentation should be delivered to the project sponsor, as a minimum.

The primary techniques you use in this process during Project Planning are stakeholder analysis, risk assessment, interviewing, negotiation, and presentation. Use these techniques to discover and cater to all factors which could ultimately jeopardize the project's success.

Work Management

The bid manager should be able to provide you with a preliminary Workplan for the project, prepared in conjunction with the proposal. Be sure to understand the assumptions and constraints upon which it was prepared. While reviewing the preliminary Workplan, it is important to remember that the planning focus for bids and project execution are different:

- The primary objective of work planning in a project bid is to determine overall project timeline and costs that are required to do the job, and therefore whether the project is feasible.
- Project work planning is focused on an efficient organization of tasks and firm control of the project.

Develop High-Level Plans and Budgets

Remember that Project Planning involves planning at a high level, for the entire duration of the project. The Workplan you develop will normally not include the names of specific persons, but it will provide realistic budgets for each role assigned to the project. However, detailed work planning for the initial phase of the project is usually done concurrently with development of the high level Workplan.

The key techniques you employ in this process during Project Planning are work planning, estimating, risk mitigation planning, and budgeting. These techniques support the development of the initial project Workplan and Finance Plan. Reuse of deliverables is an important asset to the project since it will reduce overall preparation duration of the documents. Project planning must also incorporate the planning for reuse of deliverables by incorporating the time required to prepare the deliverables for reuse and inclusion in the Knowledge Repository.

Start From a Method Approach

Oracle Method provides standard method approaches and estimating models used with tools available from Applied Business Technology (ABT) . Use ABT's Project Bridge Modeler as a high-level planning tool to construct work breakdown structures, estimates, and task dependency networks. Use ABT Project Workbench for task resourcing, scheduling, and task network refinement to create a complete, detailed Workplan for phase control.

References: The Oracle Method *Method Handbooks* provide method overviews and guidance on selecting a method approach.

Validate Your Plans

Use one or more estimating techniques, such as bottom-up estimating, to develop a high level project labor budget during Project Planning. Use top-down estimating to validate your bottom-up estimates. You should reserve an amount of effort and money as provision for each quantifiable risk, and an amount of effort and money as contingency for each risk that is not quantifiable.

Suggestion: Separate the reserves in your Workplan and Finance Plan so that transfers from the reserves to project tasks can be tracked separately.

Resource Management

Resource Management during Project Planning is somewhat unique in that you not only plan but also oversee implementation of human and physical resources at the start of the project. In addition, as in Work Management, you not only do long term planning but also plan a core set of resources needed to support execution of the first project phase. The primary techniques you use in this process during Project Planning are organizational design, recruiting, and negotiating.

Plan a Strong Project Team

You will usually work with the client project manager to plan a single project organization which includes both consulting and client staff. You will need to negotiate, both with the client to agree on an acceptable level of client resource commitment, and with consulting management to secure commitments for consulting staff to form your core project team.

Create a Project Orientation Guide

This establishes the expectations of the team members early in the engagement, facilitating the creation of a strong team. The impact of the policies and procedures established in this guide will impact the Finance Plan. Also, the very important process of time collection will be restated in the Project Orientation Guide so that all team members clearly understand the importance and ramifications of time collection and reporting.

You or your project administrator should prepare a Project Orientation Guide to help get new project members off to a good start when they arrive at the project. Use an example or template available from PJM.

Establish a Core Infrastructure

There are two key outcomes to focus on with regard to physical resources. First, use the Physical Resource Plan to define your project infrastructure, as well as who will be responsible for providing equipment and services, such as backup. Second, assign the task steps of designing and installing the core infrastructure to an individual or project team, and monitor progress closely. As a minimum, ensure that you have a working environment and Project Library set up during Project Planning.

☞ **Attention:** Establishing the project infrastructure can often be the first technical task executed on the project, and may require a technical architecture specialist and client support not yet available this early in the project. You can reduce risk by implementing only a core infrastructure here, and defer the more complex project environments, such as development and testing, to later project phases.

Prepare a Working Environment

The project staff should have a designated work area at the client site. Make sure that suitable workstations, phones, and office supplies will be available to each full time project member. Agree with the client on arrangements to accommodate new full and part time project staff in the work area. Having to find work areas each time a new staff member comes on to the site is a distraction that reduces work productivity.

Make sure that your client agrees to any arrangements for work to be done from off site locations.

Reserve conference room space to conduct meetings, testing, or reviews with project members. On medium to large projects, you may also need to establish a training area complete with terminals, personal computers, or printers.

Quality Management

It is important to ascertain the expectations of quality for the project and reflect those expectations in Project Planning deliverables. The Project Management Plan addresses these quality considerations by defining the strategies, standards, and procedures that will be used to manage the project. It also defines the staff resources, roles, and responsibilities relating to quality, such as for audits and approval of deliverables.

Suggestion: For small projects, the text in the Project Management Plan may be an adequate definition of Quality Management standards and procedures.

Use the Project Management Plan

The Project Management Plan you develop during Project Planning is not just for project management. Each person on the project is directly responsible for the quality of their own work. To make this process more effective, you should assign quality assurance responsibilities to specific individuals. In the Project Management Plan, you also define the measures to be taken on a project to ensure that quality objectives are met and a quality product is produced.

Assign Quality Roles

The key Quality Management roles are the project support specialist serving as a quality manager, quality auditor, and reviewer. The quality manager will likely be provided by the consulting practice on a full time or part time basis to help you prepare and review project quality arrangements, and oversee quality assignments. The quality auditor is the practice representative responsible for conducting periodic project audits.

You should assign senior project members to act as reviewers during quality reviews of project deliverables. They will have the experience in supervising work, evaluating technical work products, and leading review sessions. But give junior members of the project opportunities to contribute their views and opinions affecting quality as well. Your project's quality culture will benefit, and you will be helping to develop valuable skills at the same time.

Configuration Management

The most important technique you employ here during Project Planning is configuration definition. Using this technique, you analyze the deliverables for the project and establish the structure and organization of your project's configuration items. You will manage this structure in the Configuration Management Repository during the project to build your deliverables.

Control Deliverables Early

Set up a Configuration Management environment which supports at least the Project Library during Project Planning. If possible, select and acquire the project's Configuration Management toolset as part of the initial project infrastructure, so you can maintain control of project deliverables from the very beginning of the project.

Physical libraries which store reference and project materials are an ideal way to facilitate communication with the entire project team. It is also a good way to organize work. The physical library should be in a central area for all the project team members to access.

Estimating

The following chart indicates the typical percentage of effort for each task. Time not included in the estimate is indicated by an asterisk (*).

Project Planning

ID	TASK	Category Effort	Bid Manager	Client Project Manager	Consulting Business Manager	Project Support Specialist	Project Manager	Project Sponsor	Quality Auditor	Reviewer
		%								
Control and Reporting	***27.0%***									
CR.010	Establish Scope, Objectives, and Approach	17.2%	*	*	*	30	70	*		
CR.020	Define Control and Reporting Strategies, Standards, and Procedures	1.3%		*		70	30			
CR.030	Establish Management Plans	8.6%		*	*	25	75	*		
Work Management	***24.5%***									
WM.010	Define Work Management Strategies, Standards, and Procedures	1.3%		*		90	10			
WM.020	Establish Workplan	20.6%		*		90	10			
WM.030	Establish Finance Plan	2.6%		*	*	90	10			
Resource Management	***45.9%***									
RM.010	Define Resource Management Strategies, Standards, and Procedures	1.3%		*		70	30			
RM.020	Establish Staffing and Organization Plan	10.3%		*	*	40	60			
RM.025	Create Project Orientation Guide	10.3%		*	*	60	40			
RM.030	Implement Organization	12.0%		*		40	60			
RM.040	Establish Physical Resource Plan	10.3%		*		80	20			
RM.050	Establish Infrastructure	1.7%		*		80	20			
Quality Management	***1.3%***									
QM.010	Define Quality Management Strategies, Standards, and Procedures	1.3%		*		70	30			
Configuration Management	***1.3%***									
CM.010	Define Configuration Management Strategies, Standards, and Procedures	1.3%		*		80	20			

Scheduling

Project Planning tasks are executed during the initial project phase, prior to beginning execution tasks for the phase. They also include any phase planning effort necessary for the initial phase. They are normally on the project's critical path.

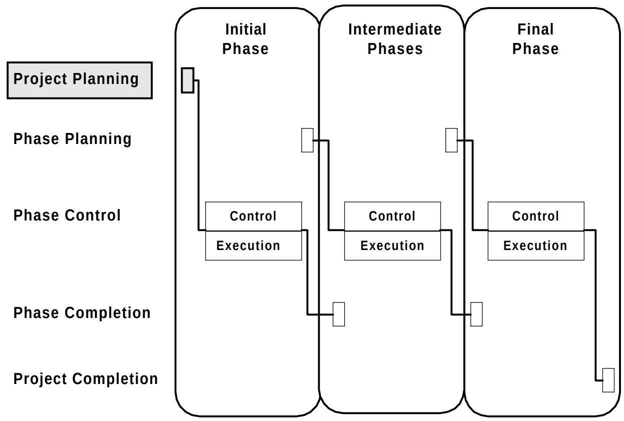

Figure 3-4 Project Planning in the Project Life-Cycle

A typical Gantt chart for Project Planning tasks is shown below. Solid bars indicate critical path tasks to complete Project Planning.

ID	Task Name
	Project Planning
CR.010	Establish Scope, Objectives, and Approach
CR.020	Define CR Strategies, Standards, and Procedures
CR.030	Establish Management Plans
WM.010	Define WM Strategies, Standards, and Procedures
WM.020	Establish Workplan
WM.030	Establish Finance Plan
RM.010	Define RM Strategies, Standards, and Procedures
RM.020	Establish Staffing and Organization Plan
RM.030	Implement Organization
RM.040	Establish Physical Resource Plan
RM.050	Establish Infrastructure
QM.010	Define QM Strategies, Standards, and Procedures
CM.010	Define CM Strategies, Standards, and Procedures

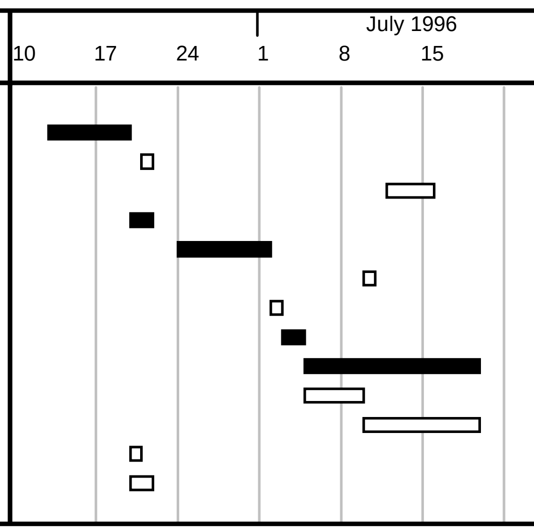

Figure 3-5 Project Planning Schedule (Scale = Weekly)

Scheduling Suggestions

Executing tasks on the critical path in parallel also allows you to compress the Project Planning timeline. The following table shows some suggested task overlaps for Project Planning.

ID	Primary Task Name	ID	Predecessor Task Name	Overlap
Work Management				
WM.010	Define Work Management Strategies, Standards and Procedures	WM.020	Establish Workplan	50%
WM.020	Establish Workplan	WM.030	Establish Finance Plan	50%
WM.020	Establish Workplan	RM.020	Establish Staffing and Organization Plan	50%
Resource Management				
RM.010	Define Resource Management Strategies, Standards, and Procedures	RM.020	Establish Staffing and Organization Plan	50%
RM.010	Define Resource Management Strategies, Standards, and Procedures	RM.040	Establish Physical Resource Plan	50%
RM.040	Establish Physical Resource Plan	WM.030	Establish Finance Plan	50%

Control and Reporting

The time required to establish the initial scoping Project Management Plan is highly dependent on the political and contractual situations at the start of the project. Make sure you allow sufficient time to interview all of the consulting and client stakeholders. Also consider how much time will be required for client review and approval of the document. Allow extra time if no risk assessment exists from the bid.

The Project Management Plan can be assembled from the strategies, standards, and procedures defined for each PJM process. On smaller projects, standards and procedures can be abbreviated or eliminated in deference to the content in the Project Management Plan itself. Standards and procedures can also be developed over time where they are not required in the initial project phase.

Several startup meetings may be required if the project team is large or geographically dispersed.

Work Management

Allow extra time if a preliminary Workplan was not constructed during the bid. Also plan extra time if a standard method will not be used as the basis for the project work breakdown structure. Allow time for iteration and overlap between scheduling work and verifying expected resource availability in Resource Management. Also, schedule enough time for detailed work planning for the initial phase.

Resource Management

Scheduling the implementation of the project infrastructure is dependent on the quantity and type of resources being acquired. Set up a phased schedule for acquiring the equipment needed immediately, then implement additional resources as they can become available.

Consider the time you will need to acquire and train the core project staff, as well as the staff for completing the initial phase. Any overlap between Resource Management tasks and project execution tasks needs to be carefully assessed for risk.

Quality Management

There are no significant scheduling issues for this process during Project Planning.

Configuration Management

There are no significant scheduling issues for this process during Project Planning.

CHAPTER

4

Phase Planning

This section describes the Phase Planning life-cycle category of PJM. The goal of Phase Planning is to thoroughly prepare for execution of the tasks required in the project phase.

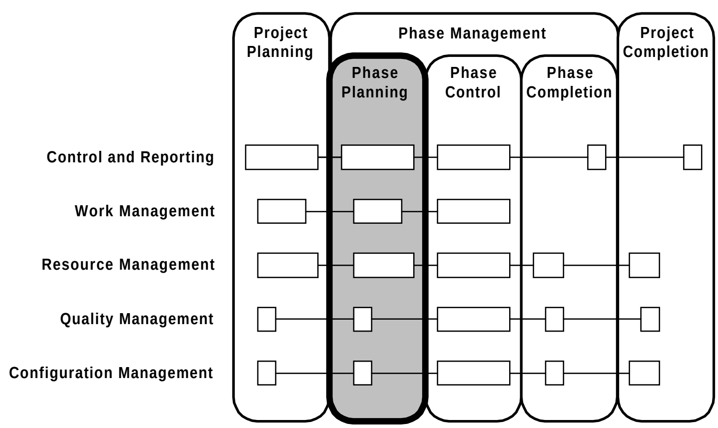

Figure 4-1 Context of PJM Phase Planning

Overview

This section provides an overview of Phase Planning. Specific topics discussed are:

- Objectives
- Critical Success Factors
- Overview Diagram
- Prerequisites
- Processes
- Key Deliverables

Objectives

The objectives of Phase Planning are:

- To update project scope to reflect agreed upon scope changes from the previous phase.
- To develop a detailed Workplan of phase execution.
- To gain commitment from the client for resources and responsibilities required from the client during the phase.
- To obtain client and consulting management approval to proceed with execution of the phase.
- To identify and establish infrastructure changes to support phase execution.
- To secure and prepare staff resources needed to execute the phase.

Critical Success Factors

The critical success factors of Phase Planning are:

- Scope changes are fully considered and accounted for in phase plans.
- New risks are identified and containment measures, provisions, or contingencies are put in place.
- Standards and procedures necessary to execute the phase and assure quality deliverables are understood by the phase staff.
- Infrastructure is installed in time for phase execution needs.
- Continuity of key personnel is secured from consulting and the client.
- Each project member understands their role in the phase Workplan and the phase's key deliverables and execution approach.

Overview Diagram

This diagram illustrates the prerequisites, processes, and key deliverables for Phase Planning.

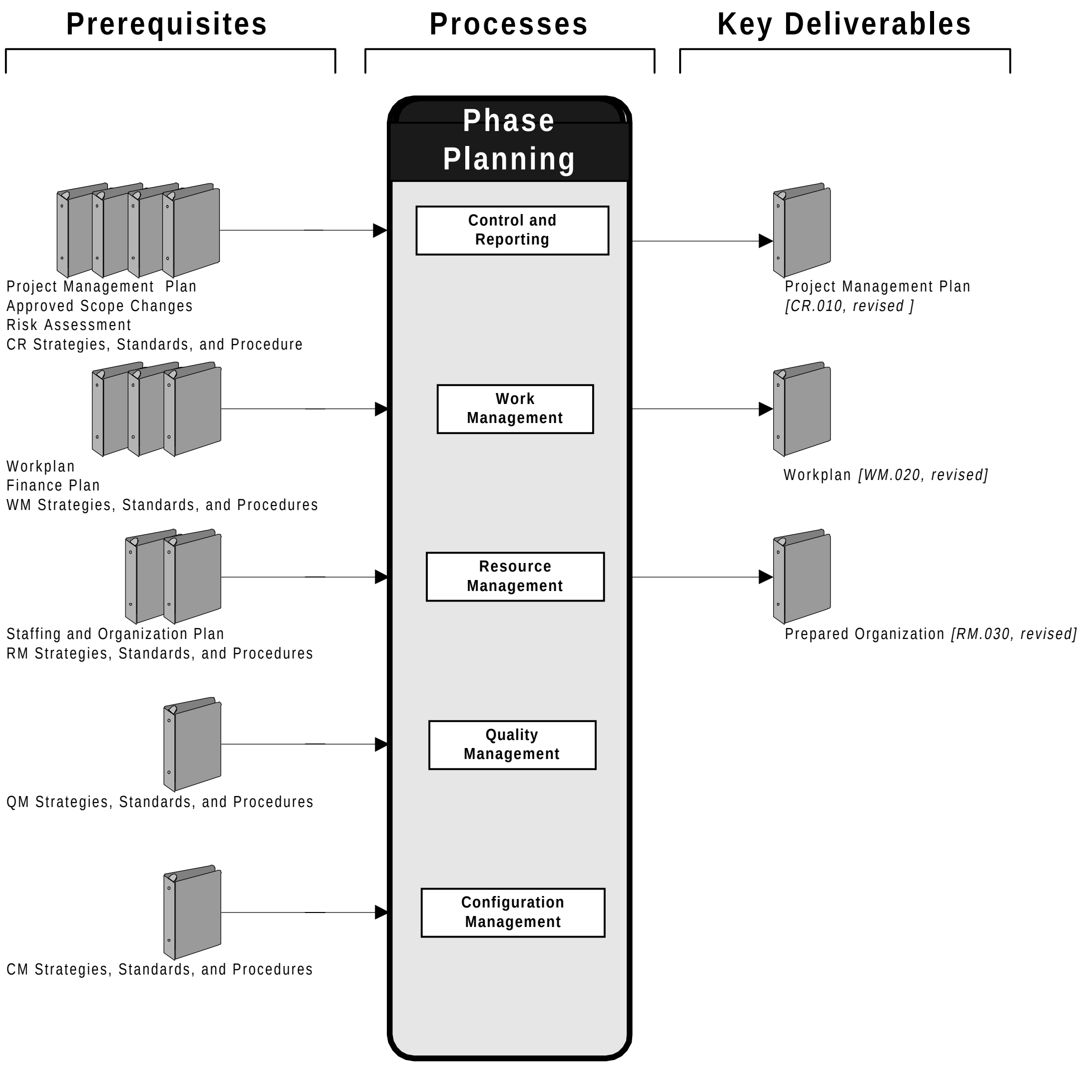

Figure 4-2 Phase Planning Overview

Prerequisites

The prerequisites for the Phase Planning phase follow. These items should exist prior to the start of the second and each subsequent phase of the project.

Prerequisite	Source
Project Management Plan	Control and Reporting
Approved Scope Changes	Control and Reporting
Risk Assessment	Control and Reporting
CR Strategies, Standards, and Procedures	Control and Reporting
Workplan	Work Management
Finance Plan	Work Management
WM Strategies, Standards, and Procedures	Work Management
Staffing and Organization Plan	Resource Management
RM Strategies, Standards, and Procedures	Resource Management
QM Strategies, Standards, and Procedures	Quality Management
CM Strategies, Standards, and Procedures	Configuration Management

Processes

The processes used in this phase follow:

Process	Description
Control and Reporting	Confirm project scope with the client and update management plans.
Work Management	Plan phase work in detail and update the Finance Plan.
Resource Management	Secure and train phase staff, set up phase organization and modify infrastructure to support phase execution.
Quality Management	Plan quality audits, healthchecks, and reviews for the phase.
Configuration Management	Plan configuration definition in detail and define the CM environment for the phase.

Key Deliverables

The key deliverables of Phase Planning are:

Deliverable	Description
Project Management Plan *[CR.010, revised]*	Revised to describe the manner in which the phase will be managed and completed.
Workplan *[WM.020, revised]*	Provides a revised detailed schedule by resource to control phase tasks, and updates the forecasts for the remaining phases of the project.
Prepared Organization *[RM.030, revised]*	Project members, organized and trained to execute the phase, with assigned roles and responsibilities.

Approach

This section describes the approach for Phase Planning. Specific topics discussed are:

- Tasks and Deliverables
- Task Dependencies
- Managing Risks
- Tips and Techniques
- Estimating
- Scheduling

Tasks and Deliverables

This table lists the tasks executed and the deliverables produced during Phase Planning. These iterated tasks are executed during each phase of the project. Processes are indicated by shaded bars.

ID	Tailored Task	Deliverable	Type
Control and Reporting			
CR.010	Establish Scope, Objectives, and Approach	Scoping Project Management Plan *[CR.010, revised]*	IT
CR.020	Define Control and Reporting Strategies, Standards, and Procedures	Control and Reporting Strategies, Standards, and Procedures *[CR.020, revised]*	IT
CR.030	Establish Management Plans	Project Management Plan *[CR.010, revised]*	IT
Work Management			
WM.010	Define Work Management Strategies, Standards, and Procedures	Work Management Strategies, Standards, and Procedures *[WM.010, revised]*	IT
WM.020	Establish Workplan	Workplan *[WM.020, revised]*	IT
WM.030	Establish Finance Plan	Finance Plan *[WM.030, revised]*	IT

ID	**Tailored Task**	**Deliverable**	**Type**
Resource Management			
RM.010	Define Resource Management Strategies, Standards, and Procedures	Resource Management Strategies, Standards, and Procedures *[RM.010, revised]*	IT
RM.020	Establish Staffing and Organization Plan	Staffing and Organization Plan *[RM.020, revised]*	IT
RM.025	Create Project Orientation Guide	Project Orientation Guide *[RM.025, revised]*	IT
RM.030	Implement Organization	Prepared Organization *[RM.030, revised]*	IT
RM.040	Establish Physical Resource Plan	Physical Resource Plan *[RM.040, revised]*	IT
RM.050	Establish Infrastructure	Prepared Infrastructure *[RM.050, revised]*	IT
Quality Management			
QM.010	Define Quality Management Strategies, Standards, and Procedures	Quality Management Strategies, Standards, and Procedures *[QM.010, revised]*	IT
Configuration Management			
CM.010	Define Configuration Management Strategies, Standards, and Procedures	Configuration Management Strategies, Standards, and Procedures *[CM.010, revised]*	IT

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Task Dependencies

This diagram shows the dependencies between tasks in Phase Planning.

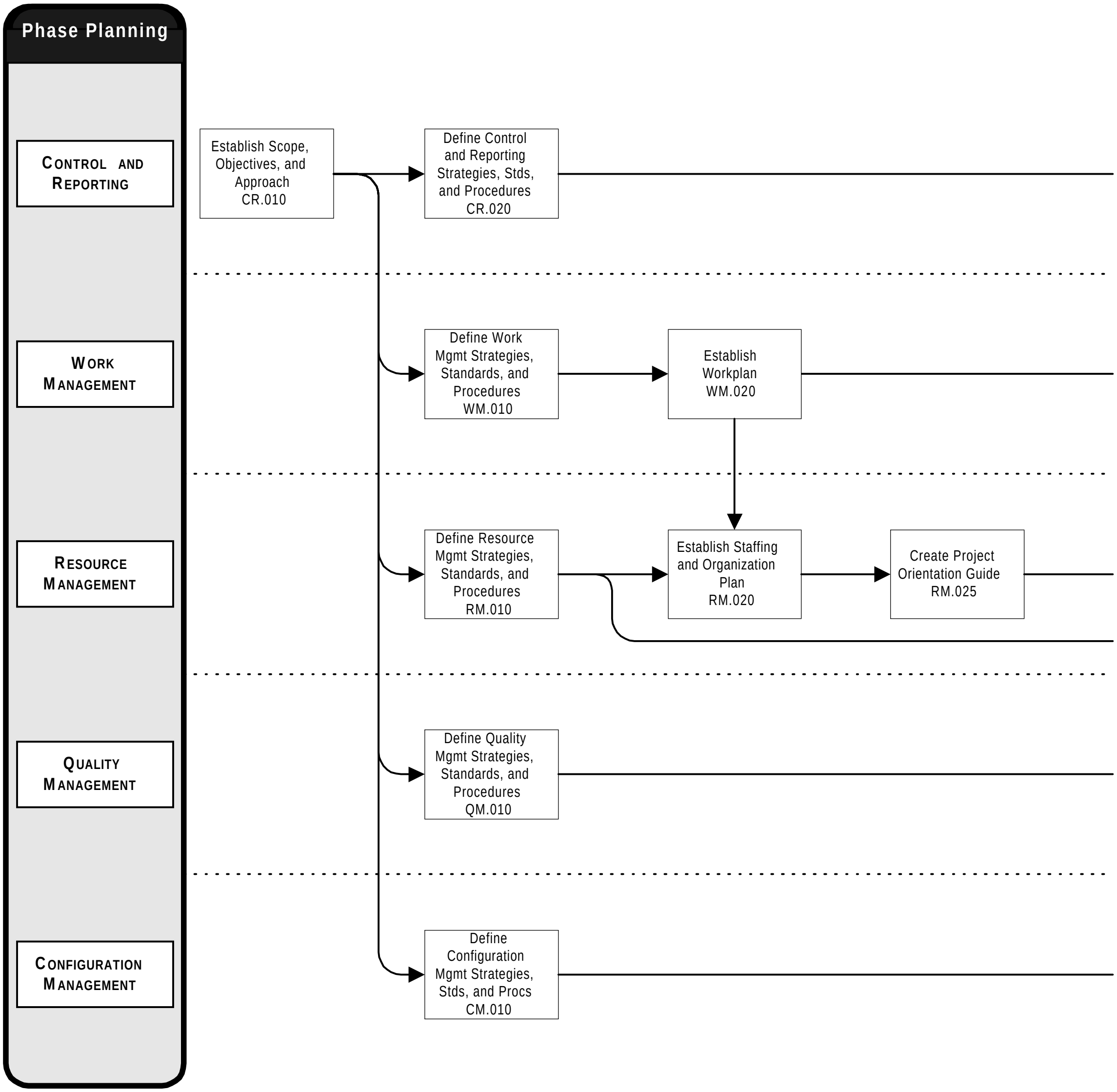

Figure 4-3 Phase Planning Task Dependencies

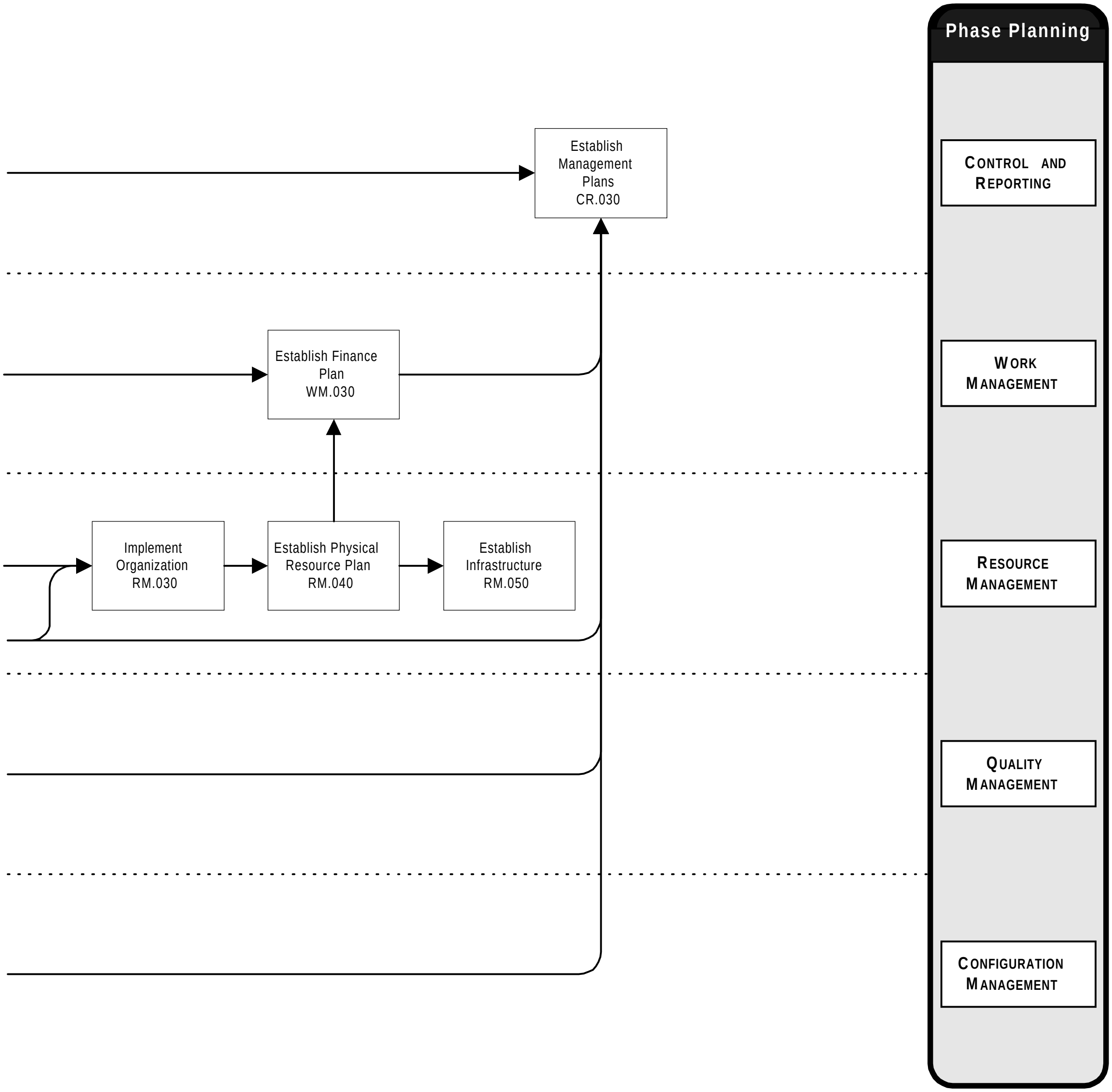

Figure 4-3 Phase Planning Task Dependencies (cont.)

Managing Risks

The most likely areas of risk during Phase Planning are the following:

- Project scope is adjusted without proper change control.
- Adequate time and resources are not committed to acquiring and training the phase staff.
- The project timeline assumes staff resources which have not been committed to the project.
- Physical resources needed during the phase are not acquired in time to support phase tasks.
- The project risk assessment is not updated prior to revising project plans for the phase.
- Acceptance procedures for phase deliverables are not agreed upon with the client.

Try to mitigate these risks in the following ways:

- Ensure that the previous phase's baseline is documented in the Project Management Plan. Secure approval of change requests for any baseline or scope changes before completing the Workplan or Finance Plan revision.
- Identify critical staff and training needs as early as possible during the prior phase and agree with consulting management on how to secure them. Assign a project management support staff member the task of revising the Staffing and Organization Plan for the phase.
- Get firm, written commitments from consulting business managers to provide staff for the phase, by name.
- Identify and order long lead-time items early. Test new software and hardware using a prototype infrastructure.
- Conduct a formal risk assessment with the client and consulting management, and explicitly factor contingencies and provisions into the Workplan and Finance Plan.
- Review planned project deliverables with the client and revise the project acceptance arrangements in the Project Management Plan.

Tips and Techniques

This section discusses techniques that may be helpful in conducting Phase Planning. It includes advice and comments on each process.

Control and Reporting

The Control and Reporting emphasis during Phase Planning should be on validating business requirements, setting expectations for the next phase, and updating assumptions and constraints. Working with the client project manager, repeat a complete but scaled-down version of Project Planning tasks, substituting reviews for development of new material. Use these tasks to also uncover changes to the project climate which could jeopardize its success. The primary Control and Reporting techniques you use during Phase Planning are risk assessment, negotiation, and presentation.

Reinforce Continuously

At the beginning of each phase, project managers reinforce client expectations by reviewing the tasks, activities, and techniques used in that phase. The client has an opportunity to choose preferred techniques, make changes to the method or approach, and influence the detailed project plan.

Confirm Client Expectations

Use phase startup meetings to keep the project focused on the business needs the project was undertaken to address. Deliver one or more presentations to communicate phase plans and validate project goals with client stakeholders. But be sensitive to the political (for example, contractual or financial) climate in the current phase, which is likely not yet complete. For example, if acceptance of current phase deliverables is an issue, consider deferral of the client startup meeting if it would help resolve current issues without undue risk. Like the project approach, these presentations are designed to be tailored to the size and needs of the project. The main goal is to set expectations during strategic points in the project.

Communicate Plans

In addition to client presentations, make introductory presentations to the entire project team on what should happen in the next phase of the project. An introduction to the phase provides the staff with specific direction. The staff gains an understanding of the process and receives instructions for conducting tasks and producing deliverables.

Set staff expectations during these introductory presentations, and communicate the following:

- phase process overview
- tasks and activities
- techniques
- deliverable tools
- control and communication tools
- events
- milestones
- schedule and resource changes
- next steps

Work Management

Planning work in the next phase is the primary focus of Phase Planning. During Phase Planning, you develop a detailed Workplan for executing the next phase, while maintaining a consistent view of the overall project plan.

Using the current Workplan as a baseline, plan the next phase in detail to create a revised Workplan. During the next phase, the revised Workplan you prepare will be used regularly by you and your project members to control, monitor, report, and modify the plan the progress of project work.

Reinforce Scope, Objectives, and Approach

Make sure that consulting and client management have a clear understanding and approve of your Work Management plans. Your Workplan will demonstrate to the client the thoroughness of your planning and preparation, and provide tangible verification of the project Scope, Objectives, and Approach. Your client will likely have responsibility for resources or tasks on which your Workplan depends. The approval of the Workplan ensures that you have commitments from your client for this support.

Plan for Control

The key techniques you employ in this process during Phase Planning are resource planning and scheduling. You may also need to add additional levels to the WBS to suit your project organization and approach.

Warning: Avoid creating Workplans with too much detail. If you need to spend more than 25% of your time updating and distributing a plan, you have defined too much detail that no one can read or understand. Instead, create a plan that tracks progress but does not micro-manage people and task steps.

Schedule Tasks Realistically

Coordinate scheduling of tasks closely with your staff resource availability. Review the staffing plan and determine expected or actual staff availability. Validate and adjust the staffing plan based on the project timeline, resource requirements profile, and staff availability. Generate and adjust the work schedule until a best schedule solution is achieved which conforms to available staff and client resources.

Use a combination of two techniques to adjust the work schedule. You can adjust (fix) the start and finish dates, or alternately, the duration of tasks, providing you abide by dependency constraints. You can also set resource availability levels for each resource to adjust the schedule, constrained by those levels.

Attention: Be sure to apply provisions for quantified risks, and contingencies for risks that are not quantifiable, to your phase Workplan based on risk mitigation strategies from Control and Reporting.

Resource Management

In Resource Management during Phase Planning, you focus on acquiring or developing a skilled project staff for executing phase tasks. In addition, you update your project infrastructure to accommodate the demands of phase tasks and staffing. However, it is likely that staff turnover and differences in skill requirements between phases should place most of your Resource Management attention on staff planning. The key staff planning techniques you will use in this process are organizational planning, skills assessment, recruitment, and negotiation.

Build a Workable Staffing Plan

Prepare a staffing plan based on the Resource Requirements Profile from Work Management. You will use this role and grade-based profile to help guide you in designing a project organization for the next phase. You will need to consider factors such as the consulting and client mix, technology specialist needs, team structure, and staffing policies from the client and consulting.

Warning: There should be a full time project manager from consulting and the client assigned to the project. Where this is not the case, or you have limited responsibility, it becomes more important to clearly define roles and responsibilities to ensure proper ownership and coordination by the client participants on the project.

Maintain Resource Continuity

Plan for resource continuity and minimized staff turnover during the phase as much as possible. Work closely with the client project manager to identify positions to be filled by client staff. Make sure that you and the client project manager understand and agree on responsibility for managing these individuals. Also, identify those individuals from your staff whose performance has particularly pleased the client.

Warning: The client may strongly request a specific consultant by name. This is a favorable situation in the sense that the client is pleased with that person’s performance and provides excellent references, albeit for a specific consultant. On the other hand, if the client ties the consultant’s involvement to the success of the project, you should intervene to prevent the client from becoming too reliant on that individual’s presence on the project.

Prepare the Phase Team

In the ideal project, staff resources are made available from the client and consulting at just the right time and with the right skills needed for the phase you are about to begin. However, in reality you will have to deal with changes in individuals’ commitments, external competition for project members, and other organizational and time constraints. Selecting, organizing, and training the project team for the phase can be one of your greatest project challenges, not to mention risks.

Prepare a Staff Training Plan listing the training each project member needs in order to fulfill their assigned roles and responsibilities. On small projects, use a simple worksheet and an adjustment to the project schedule. On larger projects, have your project coordinator help you plan and schedule training for each member using the project WBS and Workplan to properly account for resource availability.

After assessing the training needs of the project team, there are two basic options to deliver training: develop a custom curriculum that umbrellas the broad education courses identified or use the prepared education courses from educational service providers. Onsite training will often be the most timely and cost effective solution for classes larger than about eight people. Use Oracle Education Centers for smaller numbers of attendees or project members who missed onsite training.

Quality Management

During Phase Planning, ensure that project standards and procedures are updated to cover the work your project will undertake in the upcoming phase. Schedule key reviews, audits, and measurements.

Integrate Quality Measures into Plans

When you plan quality measures for the next phase, start by looking at the key deliverables and milestones for the phase. Plan quality reviews for each key deliverable your project will produce. Add additional quality reviews prior to project milestones that represent acceptance of deliverables by the client, during both execution and completion of the phase. These reviews will provide a means for the client to raise problems and your project team to remedy these problems before formal client approval.

Schedule any Quality Audits with your consulting practice that will occur during the phase. Also, select the tasks in which you will measure time, cost or quality metrics and make sure that your project procedures will support their collection.

Configuration Management

During Phase Planning, emphasize planning the CM environment to support phase execution. The CM environment consists of all of the hardware, software, tools, standards, and procedures your project uses to perform Configuration Management.

Integrate Environments

The Configuration Manager should review the environments your project will use to produce deliverables during this phase, and determine how to best integrate the CM environment with them. The results of this planning should influence how the Project Infrastructure is revised for the phase. In addition, the Configuration Manager should update the CM standards and procedures to be consistent with the phase.

Estimating

The following chart indicates the typical percentage of effort for each task. Time not included in the estimate is indicated by an asterisk (*).

Phase Planning

ID	TASK	Category Effort %	Bid Manager	Client Project Manager	Consulting Business Manager	Project Support Specialist	Project Manager	Project Sponsor	Quality Auditor	Reviewer
Control and Reporting		***19.1%***								
CR.010	Establish Scope, Objectives, and Approach	8.5%	*	*	*	30	70	*		
CR.020	Define Control and Reporting Strategies, Standards, and Procedures	1.4%		*		70	30			
CR.030	Establish Management Plans	9.2%		*	*	25	75	*		
Work Management		***47.3%***								
WM.010	Define Work Management Strategies, Standards, and Procedures	1.4%		*		90	10			
WM.020	Establish Workplan	42.4%		*		90	10			
WM.030	Establish Finance Plan	3.5%		*	*	90	10			
Resource Management		***30.7%***								
RM.010	Define Resource Management Strategies, Standards, and Procedures	1.4%		*		70	30			
RM.020	Establish Staffing and Organization Plan	3.5%		*	*	40	60			
RM.025	Create Project Orientation Guide	17.0%		*	*	60	40			
RM.030	Implement Organization	5.3%		*		40	60			
RM.040	Establish Physical Resource Plan	1.8%		*		80	20			
RM.050	Establish Infrastructure	1.8%		*		80	20			
Quality Management		***1.4%***								
QM.010	Define Quality Management Strategies, Standards, and Procedures	1.4%		*		70	30			
Configuration Management		***1.4%***								
CM.010	Define Configuration Management Strategies, Standards, and Procedures	1.4%		*		80	20			

Scheduling

Phase Planning tasks are executed during the phase prior to the phase being planned. Start Phase Planning early enough to prevent it from being on the project's critical path. Phase Planning for the initial phase is covered by the Project Planning tasks.

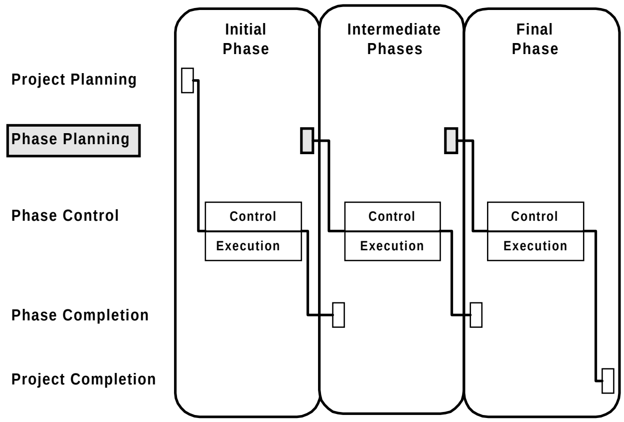

Figure 4-4 Phase Planning in the Project Life-Cycle

A typical Gantt chart for Phase Planning tasks is shown below. Solid bars indicate critical path tasks to complete Phase Planning.

ID	Task Name
	Phase Planning
CR.010	Revise Scope, Objectives, and Approach
CR.020	Revise CR Strategies, Standards, and Procedures
CR.030	Revise Management Plans
WM.010	Revise WM Strategies, Standards, and Procedures
WM.020	Revise Workplan
WM.030	Revise Finance Plan
RM.010	Revise RM Strategies, Standards, and Procedures
RM.020	Revise Staffing and Organization Plan
RM.030	Revise Organization
RM.040	Revise Physical Resource Plan
RM.050	Revise Infrastructure
QM.010	Revise QM Strategies, Standards, and Procedures
CM.010	Revise CM Strategies, Standards, and Procedures

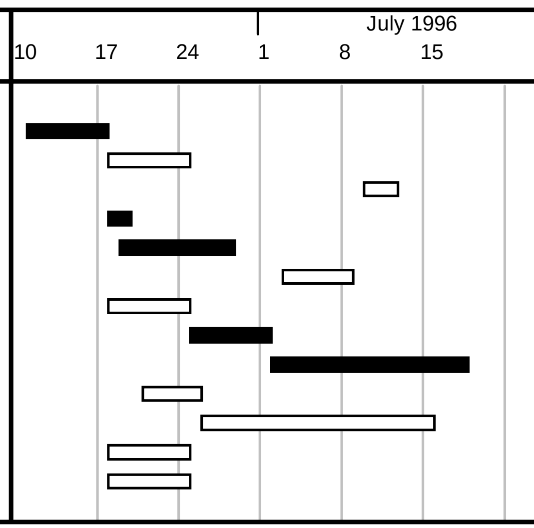

Figure 4-5 Phase Planning Schedule (Scale = Weekly)

Scheduling Suggestions

Executing tasks on the critical path in parallel also allows you to compress the Phase Planning timeline. The following table shows some suggested task overlaps for Phase Planning.

ID	Primary Task Name	ID	Predecessor Task Name	Overlap
Work Management				
WM.010	Define Work Management Strategies, Standards, and Procedures	WM.020	Establish Workplan	50%
WM.020	Establish Workplan	WM.030	Establish Finance Plan	50%
WM.020	Establish Workplan	RM.020	Establish Staffing and Organization Plan	50%
Resource Management				
RM.010	Define Resource Management Strategies, Standards, and Procedures	RM.020	Establish Staffing and Organization Plan	50%
RM.010	Define Resource Management Strategies, Standards, and Procedures	RM.040	Establish Physical Resource Plan	50%
RM.040	Establish Physical Resource Plan	WM.030	Establish Finance Plan	50%

Control and Reporting

Schedule phase startup meetings early to avoid conflicts with client or consulting schedules.

Work Management

Completing the Work Schedule is independent of having completed a sound Staffing and Organization Plan. It is acceptable to start with a preliminary version of that plan, but allow time to iterate until an acceptable compromise between time, resources, and effort is reached.

Resource Management

Acquiring and preparing the phase staff will likely be the critical path in Phase Planning. You can begin phase execution without all staff in place, but allow provisions for the risk of remaining staff not being available when expected.

Assembling staff for multiple site or geographically dispersed projects requires extra time and effort. Begin planning early with consulting practice administrators to accelerate recruiting for specific project locations.

Quality Management

There are no significant scheduling issues for this process during Phase Planning.

Configuration Management

During phases focused on documentation, CM planning is relatively simple, and focused on the Project Library. However, for phases oriented toward software deliverables, more careful integration planning will be necessary. Be sure to allow enough time during planning for these phases to include CM environment plans.

CHAPTER

5

Phase Control

This section describes the Phase Control life-cycle category of PJM. The goal of Phase Control is to monitor, measure, direct, and report on the execution of a project phase.

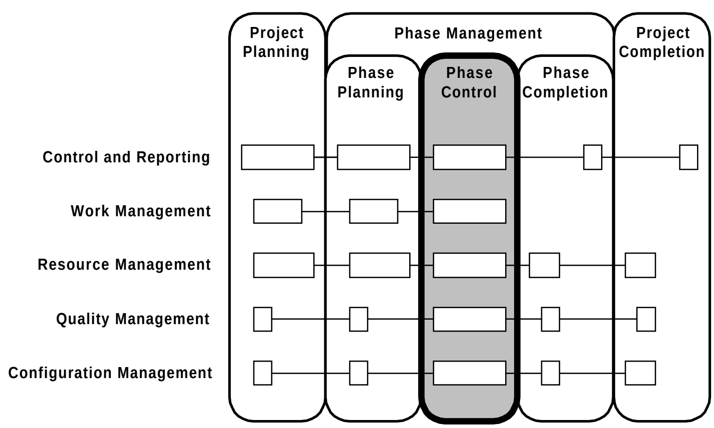

Figure 5-1 Context of PJM Phase Control

Overview

This section provides an overview of Phase Control. Specific topics discussed are:

- Objectives
- Critical Success Factors
- Overview Diagram
- Prerequisites
- Processes
- Key Deliverables

Objectives

The objectives of Phase Control are to:

- Manage the scope, quality, cost, and schedule of phase tasks and deliverables to meet or exceed client expectations.
- Compare phase execution progress to plans, identify variances, and adjust to correct significant variances.
- Anticipate possible risks to the project and take preventive measures to contain them.
- Assure the integrity of phase deliverables and that changes to them are consistent with project objectives.
- Efficiently resolve issues and problems as they are identified, identify root causes, and take corrective actions.
- Develop and direct phase staff to achieve a rewarding work environment and efficient, motivated phase organization.

Critical Success Factors

The critical success factors of Phase Control are:

- Scope, objectives, and progress of the phase are agreed on and understood by all parties.
- Commitment from project stakeholders to the project's objectives is maintained.
- Scope changes and key issues are escalated and resolved quickly and judiciously.
- Work progress is reviewed regularly and aggressive action is taken to correct variances to the Workplan.
- Project management supports and enforces quality measures.
- Project deliverables are protected from unauthorized change and baselines are fully compliant with project requirements.

Overview Diagram

This diagram illustrates the prerequisites, processes, and key deliverables for Phase Control.

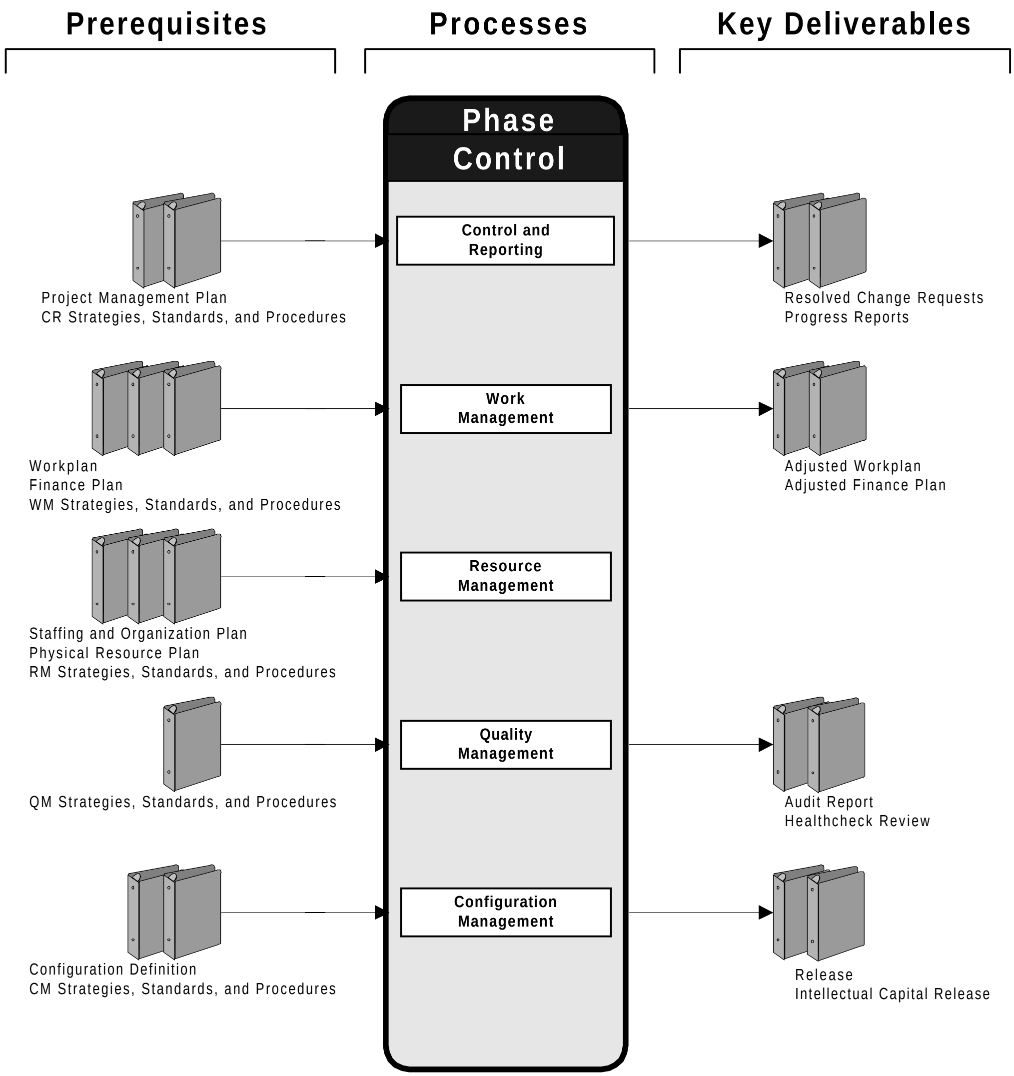

Figure 5-2 Phase Control Overview

Prerequisites

Prerequisites for Phase Control are listed below. These management documents are produced by Phase Planning to define how the phase will be executed and controlled.

Prerequisite	Source
Project Management Plan	Control and Reporting
Control and Reporting Strategies, Standards, and Procedures	Control and Reporting
Workplan	Work Management
Finance Plan	Work Management
Work Management Strategies, Standards, and Procedures	Work Management
Staffing and Organization Plan	Resource Management
Physical Resource Plan	Resource Management
Project Orientation Guide	Resource Management
Quality Management Strategies, Standards, and Procedures	Quality Management
Configuration Definition	Configuration Management
Configuration Management Strategies, Standards, and Procedures	Configuration Management

Processes

The processes used in this phase are:

Process	Description
Control and Reporting	Control risks, issues, problems, and changes. Monitor and report progress of phase.
Work Management	Control progress to Workplan and Finance Plan. Adjust Workplan to account for changes in resources, schedule, scope, and risk measures.
Resource Management	Motivate and maintain a capable Phase Organization, maintain Phase Infrastructure.
Quality Management	Review deliverables, audit processes, and monitor quality metrics.
Configuration Management	Maintain documents, control configuration items, assign baselines, and deliver releases.

Key Deliverables

The key deliverables of Phase Control are:

Deliverable	Description
Resolved Change Requests	Documented and approved changes to project scope and baseline.
Progress Reports	Status of the phase and project for communications inside and outside the project. They describe progress in terms of task completion, milestones achieved, summary of risk, issues and problems.
Adjusted Workplan	Ongoing updates to the Workplan which contain the current work status in detail and the current plan to complete the phase and project.
Adjusted Finance Plan	Ongoing updates to the Finance Plan which contain current financial status in detail and the current financial estimate to complete.
Audit Report	Documented project audit findings.
Healthcheck Review	An evaluation of the status of a project, including progress against objectives, project management, and commercial control from both consulting and client viewpoints.
Release	A baseline of phase deliverables which is made available for phase use or delivery to the client.

Deliverable	Description
Intellectual Capital Release	A release of project deliverables to the consulting Knowledge Repository.

Approach

This section describes the approach for Phase Control. Specific topics discussed are:

- Tasks and Deliverables
- Task Dependencies
- Managing Risks
- Tips and Techniques
- Estimating
- Scheduling

Tasks and Deliverables

This table lists the tasks executed and the outputs produced during Phase Control. Processes are indicated by shaded bars. Ongoing tasks, which execute throughout the phase, produce outputs on a continuous basis. Multiply instantiated tasks are executed many times, but produce a discrete deliverable upon the completion of each task instance.

ID	Task	Deliverable or Outputs	Type
Control and Reporting			
CR.040	Issue/Risk Management	Resolved Issues, Risk Containment Measures, Change Requests	O
CR.050	Problem Management	Resolved Problems, Issues/Risks, Change Requests	O
CR.060	Change Control	Scope Changes, Approved Change Requests	O
CR.070	Status Monitoring and Reporting	Client Progress Reports, Consulting Progress Reports, Action Items, Issues/Risks	O

ID	Task	Deliverable or Outputs	Type
Work Management			
WM.040	Workplan Control	Adjusted Workplan, Work Progress Statements	O
WM.050	Financial Control	Adjusted Finance Plan, Financial Progress Statements	O
Resource Management			
RM.060	Staff Control	Performance Appraisals, Adjusted Organization, Adjusted Assignment Terms of Reference	O
RM.070	Physical Resource Control	Installation Reports, Incoming Item Records, Equipment Release Records, Fault Reports	O
Quality Management			
QM.020	Quality Review	Quality Review	MI
QM.030	Quality Audit	Quality Audit	MI
QM.040	Quality Measurement	Metrics Reports	O
QM.045	Support Healthcheck	Healthcheck Review	MI
Configuration Management			
CM.020	Document Control	Controlled Documents, Document Updates, Uncontrolled Documents	O
CM.030	Configuration Control	Configuration Items, Versions, Configurations, Promotions, Baselines, Configuration Changes	O
CM.035	Knowledge Management	Intellectual Capital Releases, Reuse Assets	O
CM.040	Release Management	Releases, Release Notes	O
CM.050	Configuration Status Accounting	Configuration Management Records, Progress Statements	O

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Task Dependencies

This diagram shows information dependencies within Phase Control.

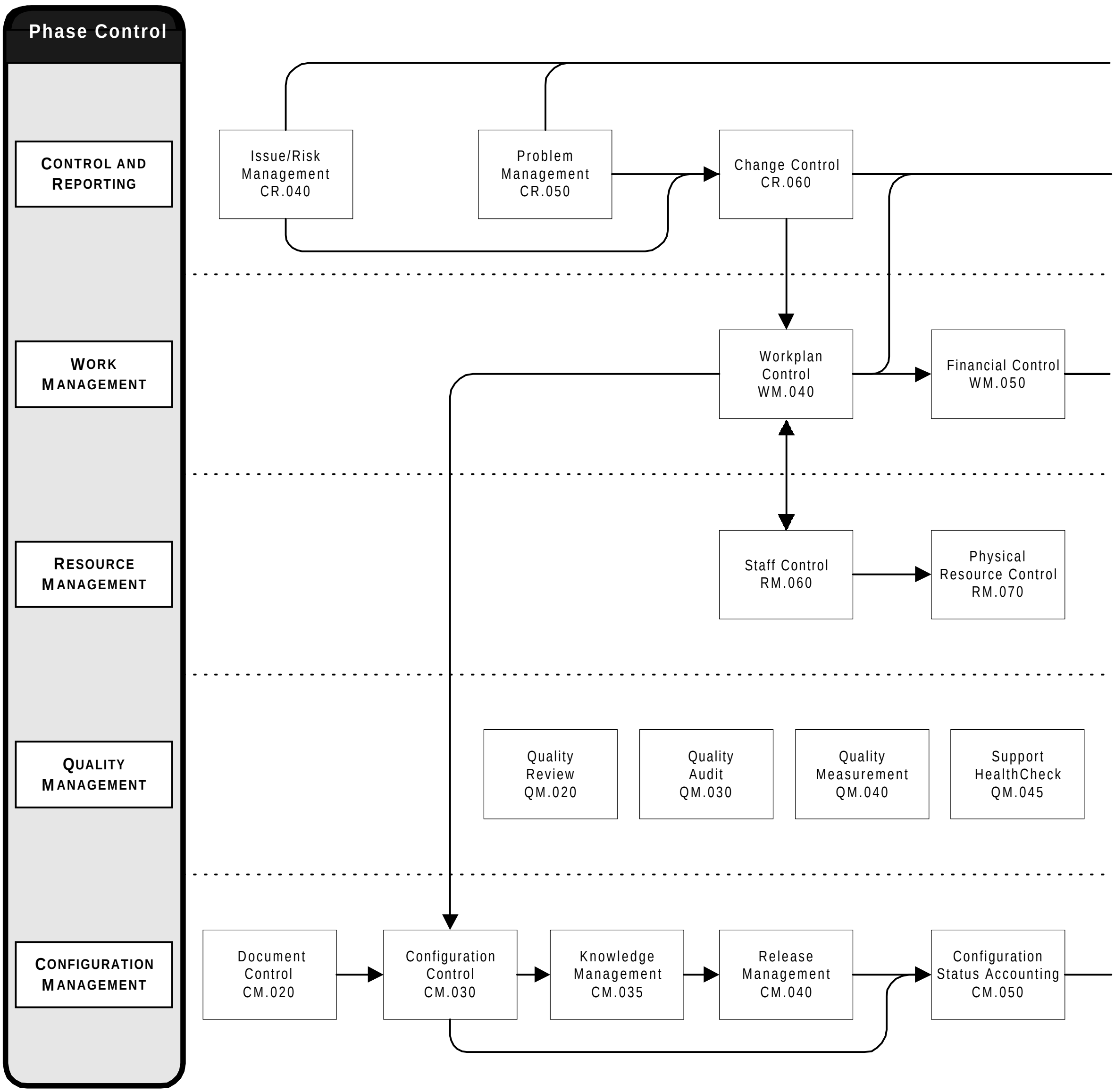

Figure 5-3 Phase Control Task Dependencies

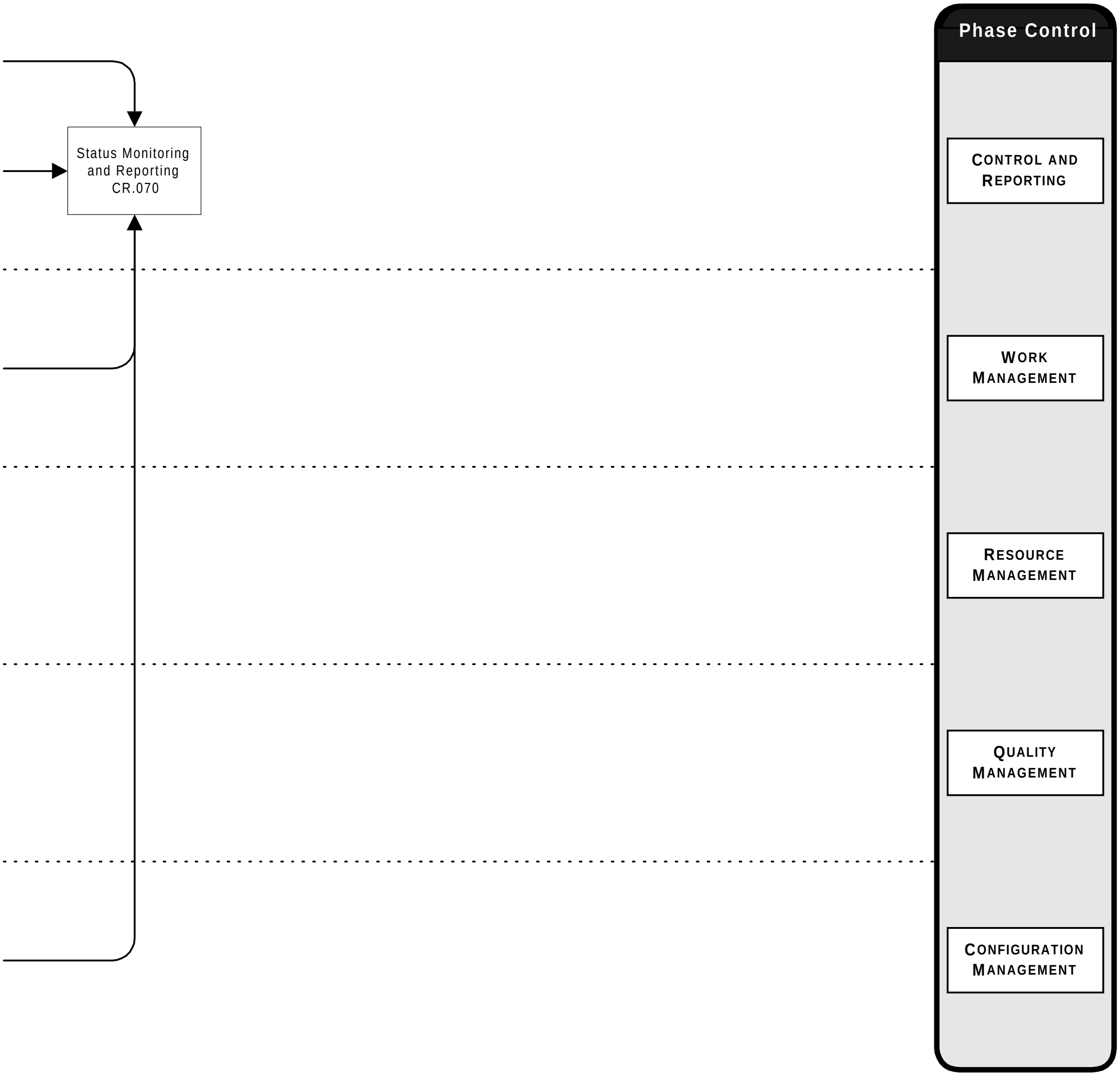

Figure 5-3 Phase Control Task Dependencies (cont.)

Managing Risks

The most likely areas of risk during Phase Control are:

- Changes are not controlled, resulting in the scope of the project gradually increasing.
- Issue, change, and problem impacts are not adequately assessed.
- Progress reviews are irregularly or superficially conducted.
- Risk mitigation strategies are not monitored for effectiveness.
- Issues are not aggressively resolved by project management.
- Deliverables are not baselined or controlled.
- Quality reviews are not conducted on key deliverables.

Try to mitigate these risks in the following ways:

- Implement and enforce the Change Control procedure on the project, and escalate changes affecting the project to the project sponsor.
- Ensure that impact assessments are thoroughly prepared and reviewed.
- Be uncompromising in insisting on a regular and complete set of project reviews, both within the project and to project stakeholders.
- Review risks regularly at project progress reviews with the client and consulting management.
- Hold regular issue review meetings and review issue backlogs at project progress reviews. Use a *top ten* issue list to maintain visibility of the most critical issues.
- Install a CM Repository, and enforce configuration management procedures.
- Schedule quality reviews for all key deliverables, and train staff in how to effectively use them.

Tips and Techniques

This section discusses techniques that may be helpful in conducting Phase Control. It includes advice and comments on each process.

Control and Reporting

You manage the tasks in this process to control the direction of the project and maintain a solid partnership with your consulting practice and the client. The key techniques you employ during Phase Control are leadership, communication, motivation, conflict management, analysis, and problem solving.

Monitor and Report Progress Regularly

The single most critical task during Phase Control is Status Monitoring and Reporting. If properly employed, the regular progress reporting and review cycles can provide a solid forum from which you can stay in control of project work. You will receive and create a wealth of qualitative and quantitative information in these reviews to help you and your project leaders find problems and fix them before they become uncontrollable.

☞ **Attention:** The following are suggested *management* reviews for your project. Make sure that you distinguish between these and project technical reviews, such as design reviews, by controlling the attendance, contributions, and agenda of the meetings. For example, a progress review attended by many users can turn into a design review if not controlled.

Team Progress Reviews

Team Progress Reviews should be held on a weekly basis to assess the progress of each team and to plan for the following week or weeks. They also include a discussion of any issues and problems. Workplans are updated in preparation for the Team Progress Review based on completion of timesheets by staff. The meeting should be chaired by the project manager.

Project Progress Reviews

Project Progress Reviews are normally held at monthly intervals, covering each financial reporting month. A report is given by the project manager to both client and consulting management, summarizing progress, problems, risks, issues, and any proposed changes. Current project status prepared by the project manager will be a key input to the meeting.

Steering Committee Meeting

The project steering committee meets at an interval usually determined by the project sponsor to review the progress reports and discuss business issues relevant to the project. The project manager, and usually the client project manager, represent the project at these meetings. The consulting business manager should be present at this review.

Warning: Never be pressured into changing estimates or direction *during* a review meeting. Take time away from the meeting environment to make a decision.

Gather Qualitative Information

The success of your project depends on a good level of communication within your project team. You must facilitate and encourage good communication by constant concern and effort. No amount of formal review can substitute for spending time informally chatting with your project members, and walking around your project work area. You will gain invaluable information about individuals, their work, and rate of progress. You will also be able to pick up on personal or political problems that can impede or are already affecting project work.

Suggestion: The qualitative information you gain, along with your own experience, should be used to corroborate the quantitative information in your project reviews. You need to feel satisfied that the figures are telling you the same thing as your impressions when you present your own progress reports.

Work Management

The Workplan Control and Financial Control tasks are usually synchronized with the Status Monitoring and Reporting task. Phase Control techniques you employ in Work Management will be a combination of those used during planning, plus performance measurement and variance analysis.

Keep the Workplan Realistic

You or your project coordinator control the Workplan for the phase by iterating through a regular tracking cycle which results in a comparison of actual progress to plan. You follow up by directing replanning to adjust the plan to reality where necessary, and by assigning corrective actions to bring future performance in conformance with the Workplan.

Warning: Watch for the following pitfalls when performing Workplan Control:

- too much replanning, insistence on a perfect Workplan
- Workplan inconsistent with the way the project is being controlled
- Workplan too general or too detailed to be useful for control
- stretching the planning baseline to match results
- project activity, but not enough tasks being completed to permit assessment of progress
- management tasks on the critical path

Good Workplan control begins with a Workplan against which project members can accurately charge their time. Each project member should submit a time record, either in the planning tool or on a separate worksheet, once each week. The total hours recorded against the project should add up to the number of hours charged to the project in the project accounting system.

Suggestion: Encourage project members to keep an ongoing record during the week of how they spent their time, rather than waiting until the day the time record is due.

Consider Using Earned Value Analysis

Earned value analysis is a technique which is commonly used to add objectivity to financial performance measurement. When you use earned value analysis, your project budget, actuals, and estimates-to-complete are tied closely to project deliverables, and can give you a more realistic view of project work. Earned value analysis involves calculating the value of a deliverable at a particular point in time, based on the budget and rules that define when the value of a deliverable is earned. You use the earned value, budget, actuals, and estimates-to-complete to provide indicators of whether or not work is being accomplished as planned.

Resource Management

During Phase Control, you manage the staff and infrastructure you have already put in place to ensure that your project can efficiently accomplish the assigned tasks. The techniques you will use most frequently during Resource Management deal with your project staff. In addition to the planning techniques you use to adjust your project organization, you will need to apply performance management, team building, motivation, leadership, and conflict management skills.

Actively Monitor Your Resources

Repeat the exercise of going through your Workplan to check which physical resources will be needed at frequent intervals. As you progress through the phase, you learn more about it, and better identify your critical resource needs. Keep careful watch over tasks which depend on a resource supplied by the client or a vendor in order to begin. Put these tasks and milestones on your Workplan to help you monitor them with the client project manager. Act decisively if a project member cites the lack of a physical resource as the reason for not completing a task on time.

Suggestion: Identify the *person* responsible for the resource and hold them to their commitment. Ensure that the person actually understands the need and is willing and able to provide you with the goods and services you expect. Be prepared to encounter dramatic examples of bureaucracy. It can require skillful negotiating and tenacity to acquire or control things over which you may have no direct authority.

Be a People Manager

As the project manager, the way you utilize your staff will have a profound impact on your project, as well as influencing their professional development. There is a substantial and dynamic body of knowledge on building, motivating, and leading project teams. Keeping current on these techniques should be a key focus area of your own professional development.

Suggestion: Do not forget yourself and your project management staff when you plan training for your project. Schedule management training or directed team building sessions using current training offerings available to you, either internally or commercially. Oracle Services offers two internal management workshops you should consider: *Introduction to Project Management in Oracle* and *Introduction to Team Leading in Oracle*.

Document the roles and responsibilities of your team members to clarify goals. If possible, integrate these assignment terms of reference with your organization’s performance management system. When you institute regular reviews of your staff’s performance, you can focus your team on project goals and also provide valuable feedback to them and the consulting practice.

The Project Orientation Guide has been developed to communicate the basic project information to the project team. Update and redistribute it to team members as the requirements change to maintain consistent performance throughout the project.

Quality Management

Quality Management tasks during Phase Control should be carefully coordinated with execution tasks. Product quality assurance techniques used during Phase Control include walk throughs, inspections, technical reviews, testing, and deliverable reviews. Process quality assurance techniques you will use include auditing, measurement, and analysis.

Enforce Quality Measures

Integrate quality measurement into every task in some way. Quality criteria are published for all Oracle Method deliverables. Use these or the PJM quality checklists to develop a culture of quality awareness on your project that places quality responsibility on your individual teams and team members. Schedule quality reviews to provide visibility and focus management attention on your phase’s key deliverables.

Warning: Follow the Project Management Plan on all levels, and do not compromise, even if the project is on a tight schedule. Once you begin to cut corners on delivering quality, you will ultimately absorb the consequences. Avoid making these types of compromises. It is important that the level of quality measures on the project be appropriate and *accounted for in the Project Management Plan*.

Improve the Process

Every quality measure should be able to communicate a message back, whether the message is positive or negative. In any case, the feedback should be constructive and informative. The recipients of the feedback should never feel like they are being policed by some governing body. Also, your feedback should never be just an identification of problems. It should always include examples, approaches, and techniques for improving the process, as implemented by the project. A Quality Plan is only effective if you can take the results and improve the process directly on your project. If you merely measure results, then you have missed the whole point of instituting quality measures.

Configuration Management

During Phase Control, you use Configuration Management tasks to protect the integrity and content of your previous phase baselines and current phase deliverables. Configuration Management is a project management process, because it implements policies you establish to safeguard your work products. On an information technology project, software often represents a large portion of the value your project delivers. Software configuration management (SCM) is especially important to project managers, because it provides visibility and organization to highly intellectual, intangible software deliverables.

Make SCM a Natural Part of Project Work

SCM controls are not meant to protect only against damage to deliverables. Some software engineers are perfectionists who will try to work in ‘one more little change’ without thinking of all of the consequences. Younger engineers may not have developed the discipline needed to coordinate their work products with those of others. Your SCM standards and procedures are sometimes there just to provide safety and to protect your project members from themselves!

Suggestion: Try to make the SCM system on your project a natural extension of your software development or implementation environment. By doing this, you can enforce SCM standards while at the same time fostering teamwork, confidence, and security.

Allow Time to Prepare Intellectual Capital

Cleansing of sensitive, proprietary, or confidential information from project deliverables may require significant effort if the deliverables are to retain the value of cumulative project experience. It is important to include an adequate amount of effort in the project workplan to support the effort of identifying what, if anything, should be edited for a larger viewing audience. As the inventory of reusable components grows, clients will benefit from the reduced effort in the earlier phases of the project since previous knowledge and deliverables will be available to offset the cleansing effort. Refer to the contractual requirements of the client prior to scheduling Knowledge Management. Schedule a knowledge review at the end of each major milestone or phase to facilitate deliverable preparation.

Attention: Scheduling and conducting a knowledge review helps the project manager facilitate gathering reusable deliverables and helps ensure that they are properly cataloged. Knowledge reviews also allow the project team to become aware of other similar projects and the potential use of intellectual capital from those projects.

Estimating

The following chart indicates the typical percentage of effort for each task. Time not included in the estimate is indicated by an asterisk (*).

Phase Control

ID	TASK	Category Effort	Bid Manager	Client Project Manager	Consulting Business Manager	Project Support Specialist	Project Manager	Project Sponsor	Quality Auditor	Reviewer
		%								
Control and Reporting	***30.8%***									
CR.040	Issue/Risk Management	6.2%		*	*	40	60			
CR.050	Problem Management	2.7%		*		50	50			
CR.060	Change Control	6.2%		*	*	60	40	*		
CR.070	Status Monitoring and Reporting	15.6%		*	*	50	50	*		
Work Management	***30.0%***									
WM.040	Workplan Control	28.1%		*		50	50			
WM.050	Financial Control	1.9%		*		70	30			
Resource Management	***6.7%***									
RM.060	Staff Control	6.2%				30	70			
RM.070	Physical Resource Control	0.5%		*		80	20			
Quality Management	***9.8%***									
QM.020	Quality Review	4.7%				60	10			30
QM.030	Quality Audit	1.7%				20	20		60	
QM.040	Quality Measurement	1.7%				60	40			
QM.045	Support Healthcheck	1.7%		*		60	40			
Configuration Management	***22.6%***									
CM.020	Document Control	2.5%		*		90	10			
CM.030	Configuration Control	6.2%		*		80	20			
CM.035	Knowledge Management	6.2%		*	*	80	20			
CM.040	Release Management	6.2%		*		90	10			
CM.050	Configuration Status Accounting	1.4%		*		80	20			

Scheduling

Phase Control tasks are executed in parallel with phase execution tasks (such as Application Implementation or Custom Development) which produce the project’s deliverable products and services.

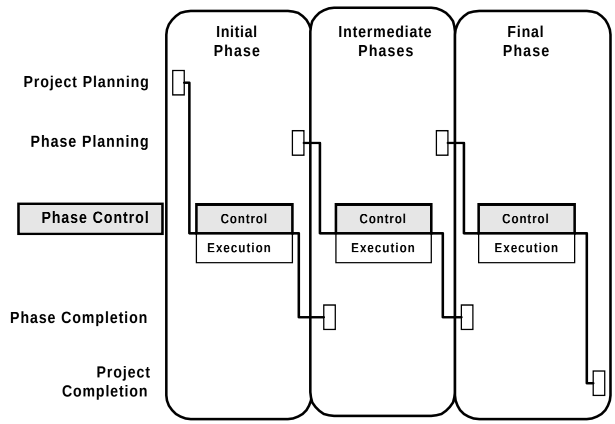

Figure 5-4 Phase Control in the Project Life-Cycle

A typical Gantt chart for Phase Control tasks is shown below.

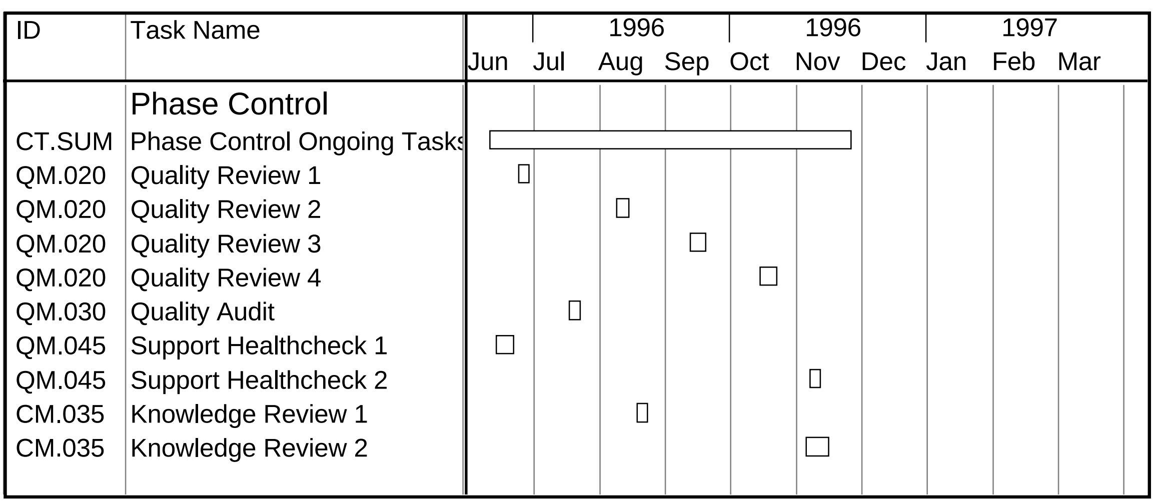

Figure 5-5 Phase Control Schedule (Scale = Monthly)

Scheduling Suggestions

Phase Control ongoing tasks overlap 100% with each other, and span the duration of phase execution. They are performed on a continuous basis, so scheduling individual tasks on the Workplan as shown above is generally unnecessary. Set up a single phase control task in the Workplan to assign resources against, if needed.

Place Phase Control multiply instantiated tasks on the Workplan as they are identified and scheduled. In addition, you should add key events such as steering committee meetings, project audits, or client reviews for management visibility.

CHAPTER

6

Phase Completion

This section describes the Phase Completion life-cycle category of PJM. The goal of Phase Completion is to secure client approval of key deliverables produced during the phase and acknowledgment that the phase has been completed to the mutual satisfaction of consulting and client.

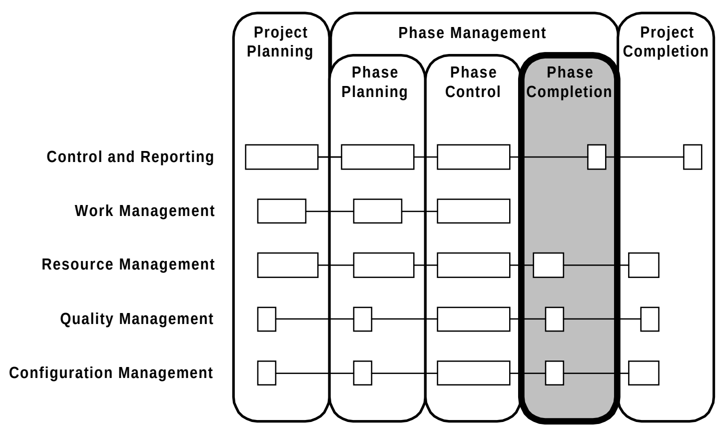

Figure 6-1 Context of PJM Phase Completion

Overview

This section provides an overview of Phase Completion. Specific topics discussed are:

- Objectives
- Critical Success Factors
- Overview Diagram
- Prerequisites
- Processes
- Key Deliverables

Objectives

The objectives of Phase Completion are to:

- Verify that phase deliverables meet project quality and completeness standards.
- Secure client acceptance of phase deliverables.
- Release staff and physical resources no longer required.

Critical Success Factors

The critical success factors of Phase Completion are:

- The phase acceptance procedure is clearly communicated in the Project Management Plan.
- Client satisfaction concerns are identified and addressed prior to requesting acceptance of deliverables.
- Outstanding issues and problems which affect phase deliverables are resolved prior to their acceptance.
- Change control, configuration control, and quality records are complete and accurate, and demonstrate compliance with project standards.

Overview Diagram

This diagram illustrates the prerequisites, processes, and key deliverables for Phase Completion.

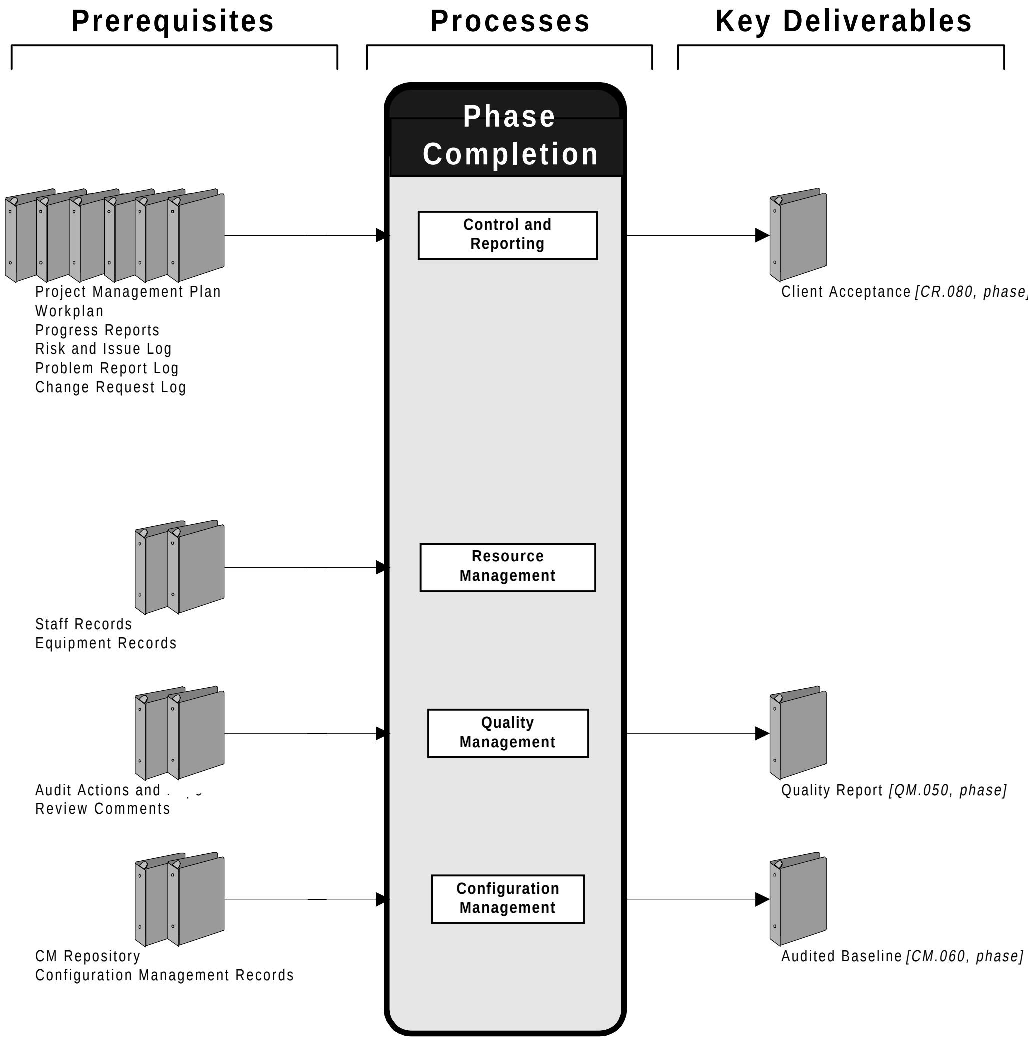

Figure 6-2 Phase Completion Overview

Prerequisites

The prerequisites for Phase Completion follow. These items are produced or revised earlier in the phase.

Prerequisite	Source
Project Management Plan	Control and Reporting
Workplan	Work Management
Progress Reports	Control and Reporting
Risk and Issue Log	Control and Reporting
Problem Report Log	Control and Reporting
Change Request Log	Control and Reporting
Staff Records	Resource Management
Equipment Records	Resource Management
Audit Actions and Reports	Quality Management
Review Comments	Quality Management
CM Repository	Configuration Management
Configuration Management Records	Configuration Management

Processes

The processes used in this phase are:

Process	Description
Control and Reporting	Obtain approval of phase deliverables to the mutual satisfaction of consulting and the client.
Resource Management	Release staff and physical resources not required for further project work.
Quality Management	Conduct an assessment of the completeness of quality control arrangements at the end of the phase.
Configuration Management	Verify that key phase deliverables are those intended and that adequate control of their development was exercised.

Key Deliverables

The key deliverables of Phase Completion are:

Deliverable	Description
Client Acceptance *[CR.080, phase]*	Client approval of phase tasks and deliverables.
Quality Report *[QM.050, phase]*	Assessment of the completeness of quality measures during the phase.
Audited Baseline *[CM.060, phase]*	Confirmed functional and physical completeness of key phase deliverables and capture of the intellectual capital.

Approach

This section describes the approach for Phase Completion. Specific topics discussed are:

- Tasks and Deliverables
- Task Dependencies
- Managing Risks
- Tips and Techniques
- Estimating
- Scheduling

Tasks and Deliverables

This table lists the tasks executed and the deliverables produced during Phase Completion. The multiply occurring tasks are repeated for each phase of the project. Processes are indicated by shaded bars.

ID	Tailored Task	Deliverable	Type
Control and Reporting			
CR.080	Secure Client Acceptance	Client Acceptance *[CR.080, phase]*	MO
Resource Management			
RM.080	Release Staff	Released Staff *[RM.080, phase]*	MO
RM.090	Release Physical Resources	Released Physical Resources *[RM.090, phase]*	MO
Quality Management			
QM.050	Perform Quality Assessment	Quality Report *[QM.050, phase]*	MO
Configuration Management			
CM.060	Audit Key Deliverables	Audited Baseline *[CM.060, phase]*	MO

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Task Dependencies

This diagram shows dependencies between tasks in Phase Completion.

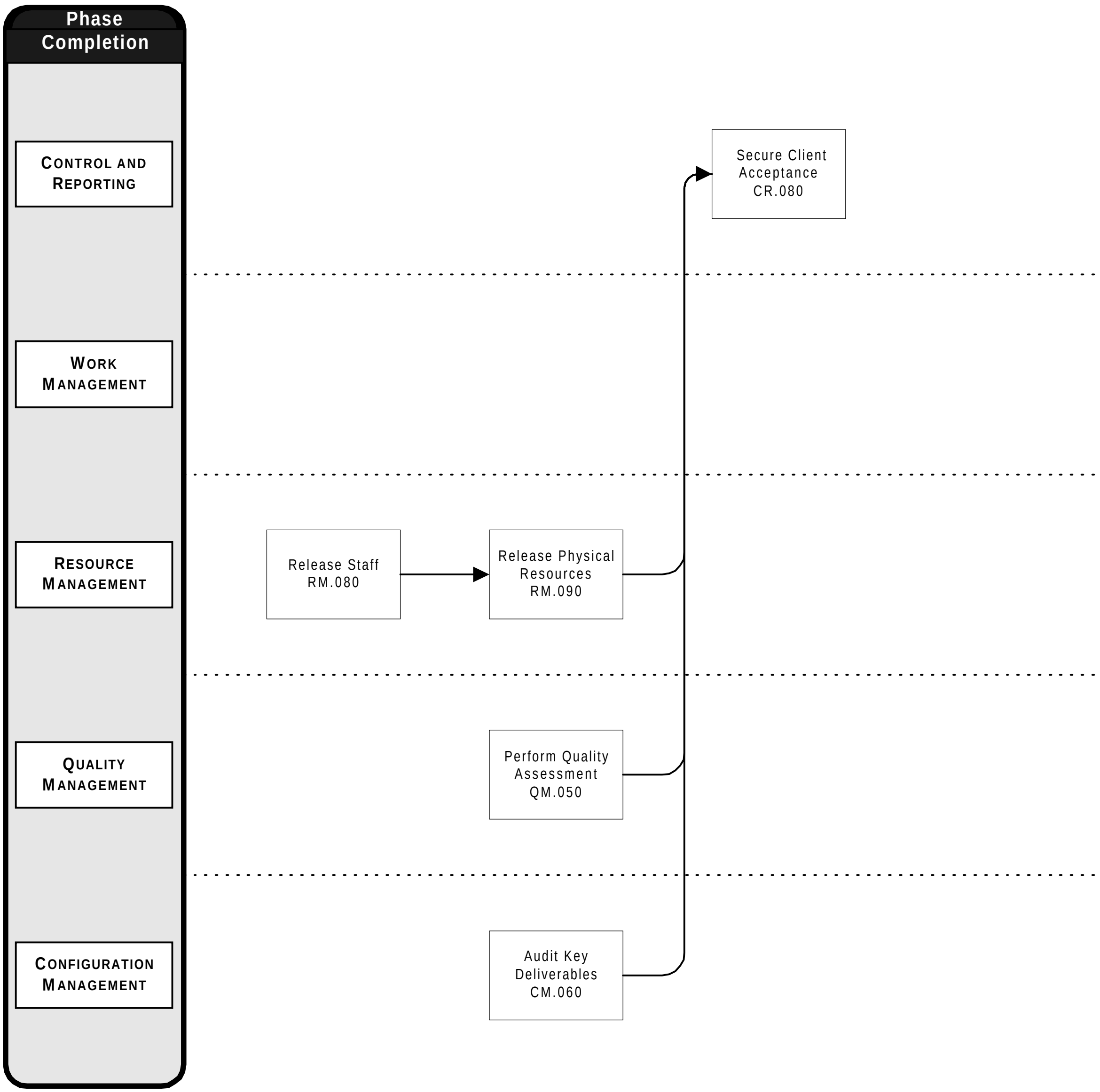

Figure 6-3 Phase Completion Task Dependencies

Managing Risks

The most likely areas of risk during Phase Completion are:

- The client is not prepared to participate in deliverable approval.
- Lingering issues or problems prevent approval.
- Change control, configuration control, or quality assurance actions were not adequately recorded, resulting in questions about the quality, completeness, or correctness of phase deliverables.
- A client stakeholder raises a *show-stopping* issue just prior to beginning phase acceptance.

Try to mitigate these risks in the following ways:

- Review the acceptance procedure with the client project manager and the client project members participating in the approval.
- Hold one or more issue and problem resolution sessions prior to beginning phase completion.
- During phase control, verify that records are adequate using one or more quality audits.
- Keep the project sponsor informed of progress during the phase, and use stakeholder analysis and communication techniques to identify new client concerns as early as possible.

Tips and Techniques

This section discusses techniques that may be helpful in conducting Phase Completion. It includes advice and comments on each process.

Control and Reporting

The focus of this process during Phase Completion is closing the project phase by obtaining client acceptance of phase deliverables. Techniques you will use are communication and presentation. Use CR.080 throughout the phase, if your plans call for incremental client acceptance as deliverables are produced. Each instance of the task produces an Acceptance Certificate documenting client acceptance of that set of deliverables. The best way to prepare for each acceptance is to use QM.020 to identify and resolve client concerns about the deliverables.

Consider a Phase End Report

For smaller projects where there are simple lines of communication and the client relationship is sound, it is more efficient to finalize all of the key phase deliverables at one time and present them in one place. A common way of accomplishing this is to produce a Phase End Report, which includes your phase's key deliverables, or documentation supporting them. It also increases consulting credibility with the client to produce a nice book to mark the end of a phase. The Phase End Report should consist of:

- a general section: standard front matter, a management summary, and phase specific content (vision, findings, recommendations, etc.).
- a detailed section: appendices with copies of key documents from the phase.

Resource Management

The release of staff and physical resources assumes you have available to you the Staffing and Organization Plan and Physical Resource Plan for the next phase. Give departing project members a final performance appraisal and exit interview before they return to their practices.

Quality Management

The Quality Report gives you an opportunity to demonstrate to the client the completeness of your quality control measures during the phase. The Quality Assessment task may be conducted by a member of the project staff or may be performed by an external quality consultant.

Configuration Management

A Phase End Release can be used as part of, or in place of, the Phase End Report discussed above. A Phase End Release consists of all key deliverables of the phase (for example, the Audited Baseline), delivered to the client in the agreed upon medium. You can find standard key deliverables listed for each phase in the applicable Method Handbook for the project approach you are using.

Estimating

The following chart indicates the typical percentage of effort for each task. Time not included in the estimate is indicated by an asterisk (*).

Phase Completion

ID	TASK	Category Effort %	Bid Manager	Client Project Manager	Consulting Business Manager	Project Support Specialist	Project Manager	Project Sponsor	Quality Auditor	Reviewer
Control and Reporting	***21.6%***									
CR.080	Secure Client Acceptance	21.6%		*		25	75	*		
Resource Management	***22.5%***									
RM.080	Release Staff	16.2%		*		20	80			
RM.090	Release Physical Resources	6.3%				90	10			
Quality Management	***30.6%***									
QM.050	Perform Quality Assessment	30.6%				95	5			
Configuration Management	***25.2%***									
CM.060	Audit Key Deliverables	25.2%				85	15			

Scheduling

Phase Completion tasks should be executed during the succeeding phase, to be off of the project's critical path. However, the scope, objectives, and approach may specifically call for completion of a phase to precede commencement of the succeeding phase. Phase Completion tasks are replaced by Project Completion tasks in the final phase.

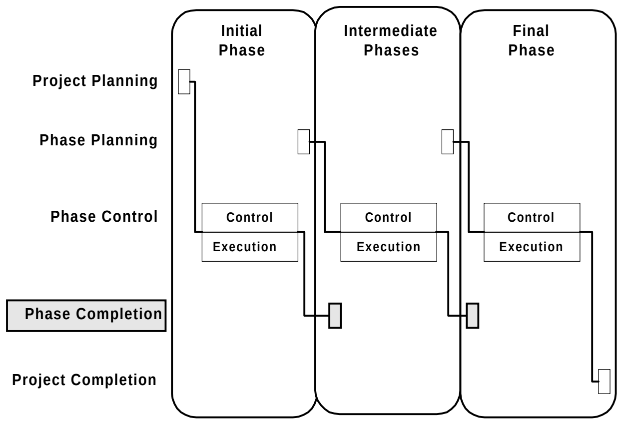

Figure 6-4 Phase Completion in the Project Life-Cycle

A typical Gantt chart for Phase Completion tasks is shown below. Solid bars indicate critical path tasks to complete Phase Completion.

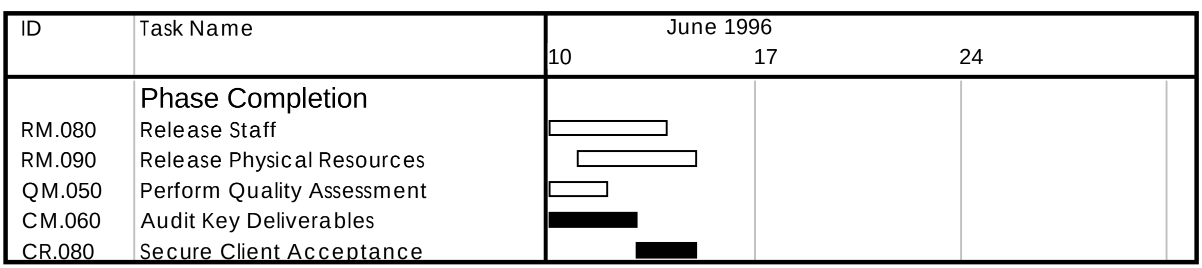

Figure 6-5 Phase Completion Schedule (Scale = Weekly)

Scheduling Suggestions

Executing tasks on the critical path in parallel also allows you to compress the Phase Completion timeline. The following table shows some suggested task overlaps for Phase Completion.

ID	Primary Task Name	ID	Predecessor Task Name	Overlap
Resource Management				
RM.080	Release Staff	RM.090	Release Physical Resources	75%
RM.090	Release Physical Resources	CR.080	Secure Client Acceptance	100%
Quality Management				
QM.050	Perform Quality Assessment	CR.080	Secure Client Acceptance	75%
Configuration Management				
CM.060	Audit Key Deliverables	CR.080	Secure Client Acceptance	75%

The task overlaps with CR.080, Secure Client Acceptance, represent the assessment of client satisfaction and resolving remaining issues and problems for the phase.

CHAPTER

7

Project Completion

This section describes the Project Completion life-cycle category of PJM. The goal of Project Completion is to secure client acceptance of the project and to close the project in an orderly fashion. A well-managed Project Completion demonstrates that you have met your client's needs and allows you to negotiate a successful conclusion to the project. A satisfied client will act as a reference and a potential source of future engagements.

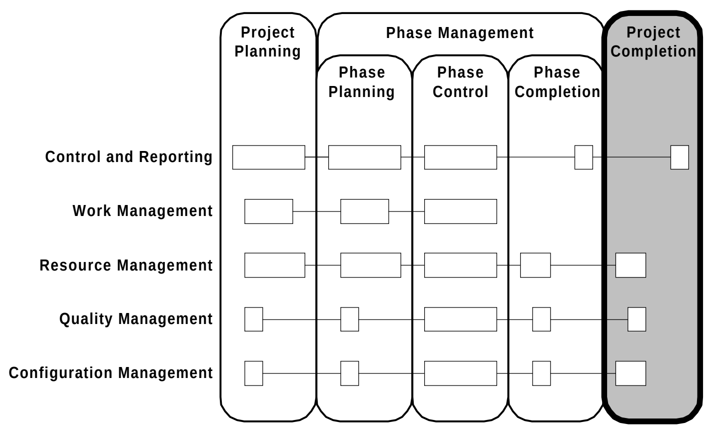

Figure 7-1 Context of PJM Project Completion

Overview

This section provides an overview of Project Completion. Specific topics discussed are:

- Objectives
- Critical Success Factors
- Overview Diagram
- Prerequisites
- Processes
- Key Deliverables

Objectives

The objectives of Project Completion are to:

- Gain client acceptance of all project deliverables.
- Close out the contractual agreement, if any, with the client.
- Hand over project deliverables and environments to the production support team (as appropriate).
- Release staff and physical resources.
- Document and archive project results in the consulting practice.

Critical Success Factors

The critical success factors of Project Completion are:

- Early attention to acceptance planning and definition of acceptance criteria.
- Good client relationship and managed expectations.
- Time allowed for rework during acceptance.
- Outstanding issues and problems resolved prior to their acceptance.

Overview Diagram

This diagram illustrates the prerequisites, processes, and key deliverables for Project Completion.

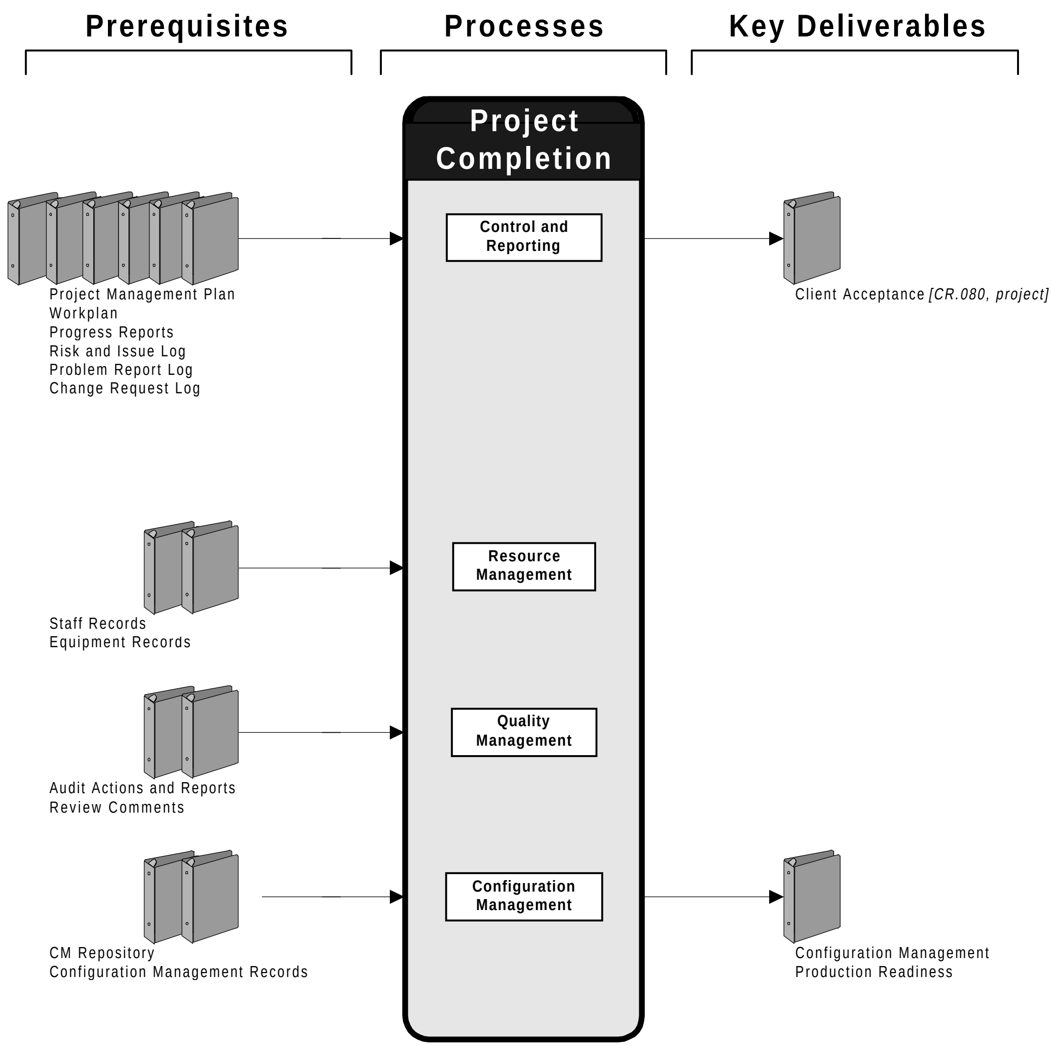

Figure 7-2 Project Completion Overview

Prerequisites

The prerequisites for Project Completion follow. These items are produced or revised earlier in the project.

Prerequisite	Source
Project Management Plan	Control and Reporting
Workplan	Work Management
Progress Reports	Control and Reporting
Risk and Issue Log	Control and Reporting
Problem Report Log	Control and Reporting
Change Request Log	Control and Reporting
Staff Records	Resource Management
Equipment Records	Resource Management
Audit Actions and Reports	Quality Management
Review Comments	Quality Management
CM Repository	Configuration Management
Configuration Management Records	Configuration Management

Processes

The processes used in this project are:

Process	Description
Control and Reporting	Obtain approval of project deliverables, assess client satisfaction, and archive project results.
Resource Management	Release all remaining staff and physical resources.
Quality Management	Support client acceptance of deliverables by assessing quality measures.
Configuration Management	Verify project deliverables and transfer the CM environment to the client, if appropriate.

Key Deliverables

The key deliverables of Project Completion are:

Deliverable	Description
Client Acceptance *[CR.080, project]*	Client approval of project deliverables, leading to contractual closure.
Configuration Management Production Readiness	The capability of the client to continue performing Configuration Management during production.

Approach

This section describes the approach for Project Completion. Specific topics discussed are:

- Tasks and Deliverables
- Task Dependencies
- Managing Risks
- Tips and Techniques
- Estimating
- Scheduling

Tasks and Deliverables

This table lists the tasks executed and the deliverables produced during Project Completion. These singly instantiated tasks are executed during Project Completion. Processes are indicated by shaded bars.

ID	Tailored Task	Deliverable	Type
Control and Reporting			
CR.080	Secure Client Acceptance	Client Acceptance *[CR.080, project]*	SI
Resource Management			
RM.080	Release Staff	Released Staff *[RM.080, project]*	SI
RM.090	Release Physical Resources	Released Physical Resources *[RM.090, project]*	SI
Quality Management			
QM.050	Perform Quality Assessment	Quality Report *[QM.050, project]*	SI
Configuration Management			
CM.060	Audit Key Deliverables	Audited Baseline *[CM.060, project]*	SI
CM.070	Conclude Configuration Management	Configuration Management Production Readines	SI

Type: SI=singly instantiated, MI=multiply instantiated, MO=multiply occurring, IT=iterated, O=ongoing. See Glossary.

Task Dependencies

This diagram shows dependencies between tasks in Project Completion.

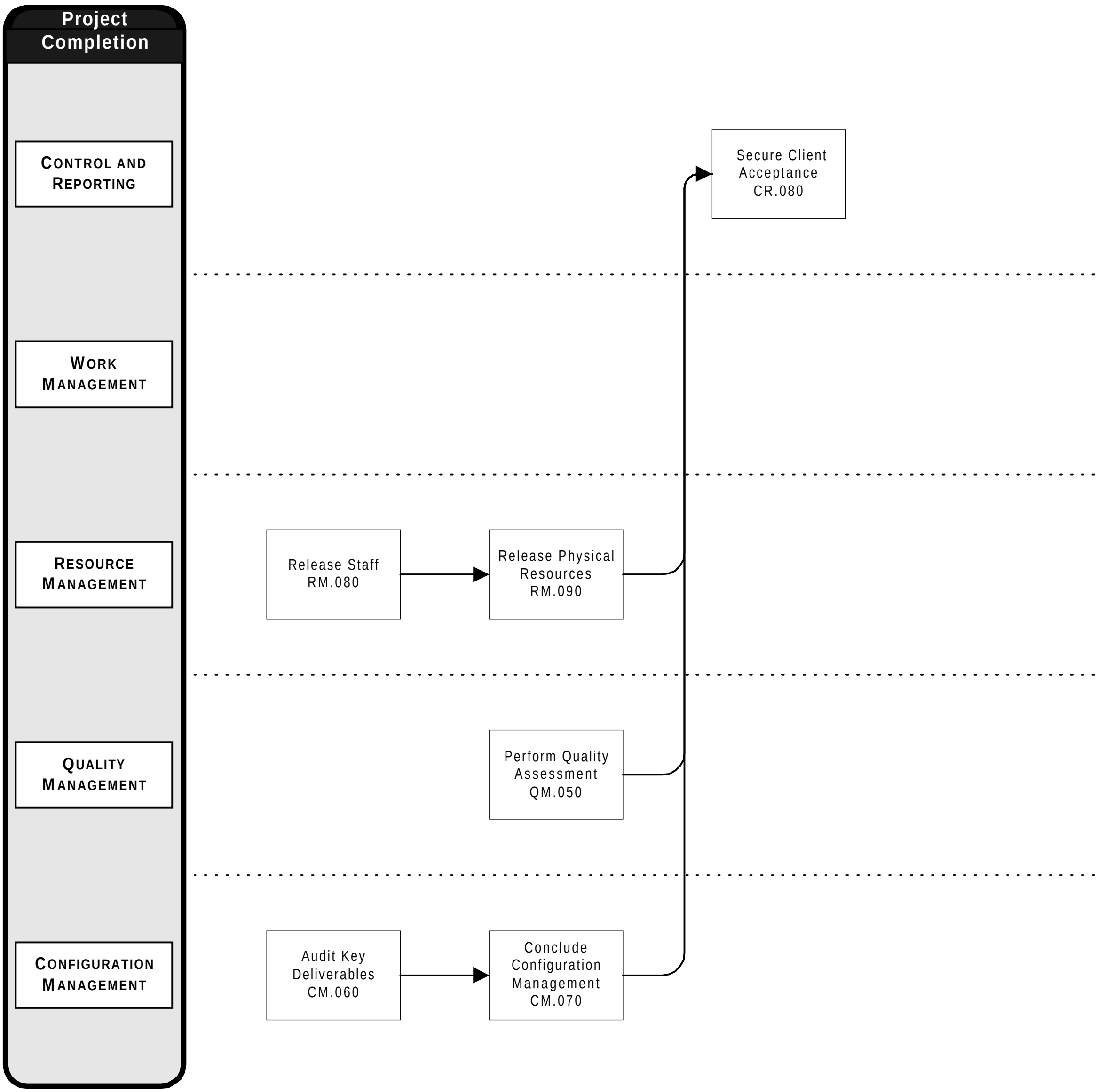

Figure 7-3 Project Completion Task Dependencies

Managing Risks

The most likely areas of risk during Project Completion are:

- Acceptance criteria and procedures not agreed upon with the client.
- Remaining questions about the quality, completeness, or correctness of project deliverables.
- The client has not internally agreed on the individuals responsible for project approval.

Try to mitigate these risks in the following ways:

- Use the Scope, Objectives, and Approach and Project Management Plan documents to agree on approval procedures during Project Planning, and confirm them during planning for the final phase.
- Aggressively resolve issues and problems during final testing of the deliverable solution.
- Request that your consulting business manager discuss client approval responsibilities with the project sponsor during preparation for acceptance.

Tips and Techniques

This section discusses techniques that may be helpful in conducting Project Completion. It includes advice and comments on each process.

Control and Reporting

This process manages the administrative and contractual closing of the project with the client and the consulting practice. Techniques you find most useful are communicating, negotiating, and conflict resolution. The general approach recommended for Project Completion is to make the client responsible for conducting acceptance tasks, such as testing. However, specific terms and conditions may be detailed in the contractual agreement and should be reflected in project management plans.

Manage Acceptance Expectations Carefully

When conducting Project Completion, remember to continue to manage changes, issues, and problems throughout acceptance. Last minute issues and problems can be quite common, as client stakeholders realize that they have only a short time remaining to influence project results. The project sponsor and client project manager should take a leadership role at this point in the project to control the introduction of new issues from the client.

Feed Back Results to Your Practice

It is easy to focus on contractual and resource issues during the final days of a project. However, do not forget that your project has advanced your consulting practice's capabilities and has hopefully produced a product that you and the client are proud of. While you have the time, staff, and client contacts available, make sure to feed project results back to your practice. Use the Project End Report to capture information about the project for marketing and to benefit future projects. The Client Satisfaction Report will provide objective information about project results to your consulting management.

Resource Management

Coordinate closely with your consulting business manager about resource needs after the project closes. There may be uncompleted negotiations regarding a support or maintenance engagement which could retain some of your project staff. Once post-project commitments to the project have been defined, staff can be reallocated. In practice, as the project is in the completion phase, then reallocation can be started.

Physical resources can easily become lost or intermixed in a project infrastructure shared with the client. If you have maintained accurate equipment records, you should be able to sort out which equipment, licenses, and materials are to be retained by you and by the client. Otherwise, you leave yourself vulnerable to a possible contractual dispute with the client.

Quality Management

At this point in the project, previous phase acceptances should have already established a clear pattern of quality compliance. Use the final Quality Report as a tool to support your final approval, or to help resolve any contractual disputes or issues regarding the quality of your project deliverables.

Configuration Management

The transfer of the Configuration Management environment to the client should be coordinated with transfers of other project environments, specifically those for development, maintenance, and testing. The amount of transfer is project-specific, and should be worked out well in advance, ideally as part of the contractual agreement.

Archive Project Deliverables

Configuration Management is also responsible for transferring archive copies of project deliverables to your consulting practice. Follow your practice policies and procedures about archiving project work products. Consider these questions before contributing your work:

- Do you have the legal right to remove deliverables? Check the contractual agreement. Also, it is good consulting practice to inform the client of your intentions and ask for permission, even if you have legal authority.
- Does the client have any legal or other reservations about using the client name in the body of the materials? If so, then you must replace that name with one that is generic so deliverables cannot be traced to the client.
- Does the deliverable require any more information in order to be valuable to the practice? If so, then add a preface to the deliverable with the proper explanation or substitute the deliverable prior to submitting.

☞ **Attention: Take legal restrictions seriously**. Clients may have confidential information that if disclosed, even as a sample, to their competitors would be detrimental to their position in the marketplace. Your practice has an obligation to protect the information of your clients. You may be placing it at legal and financial risk if you disclose client information.

Estimating

The following chart indicates the typical percentage of effort for each task. Time not included in the estimate is indicated by an asterisk (*).

Project Completion

ID	TASK	Category Effort %	Bid Manager	Client Project Manager	Consulting Business Manager	Project Support Specialist	Project Manager	Project Sponsor	Quality Auditor	Reviewer
Control and Reporting	***12.8%***									
CR.080	Secure Client Acceptance	12.8%		*		25	75	*		
Resource Management	***25.5%***									
RM.080	Release Staff	17.0%		*		20	80			
RM.090	Release Physical Resources	8.5%				90	10			
Quality Management	***17.0%***									
QM.050	Perform Quality Assessment	17.0%				95	5			
Configuration Management	***44.7%***									
CM.060	Audit Key Deliverables	14.9%				85	15			
CM.070	Conclude Configuration Management	29.8%		*		80	20			

Scheduling

Project Completion tasks are executed at the end of the final project phase. Completion of the final phase is included in the scope of Project Completion tasks.

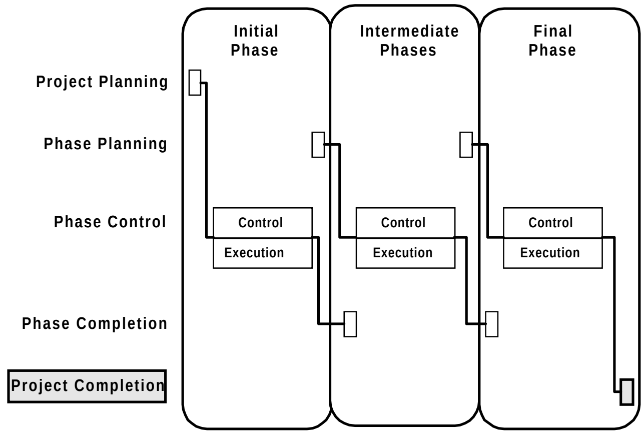

Figure 7-4 Project Completion in the Project Life-Cycle

A typical Gantt chart for Project Completion tasks is shown below. Solid bars indicate critical path tasks to complete Project Completion.

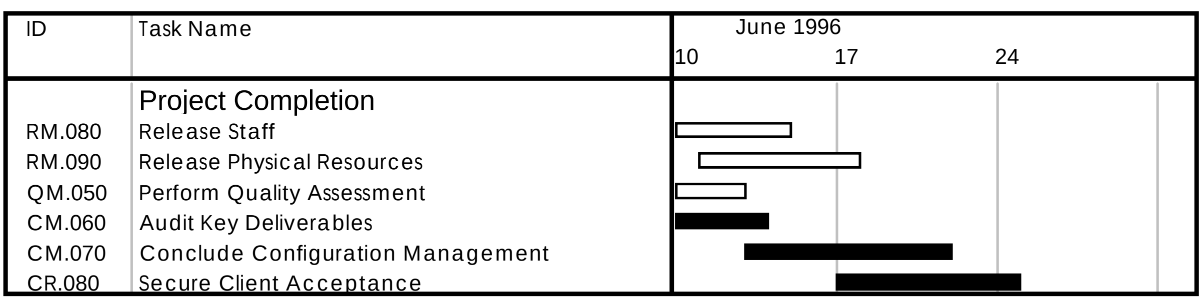

Figure 7-5 Project Completion Schedule (Scale = Weekly)

Scheduling Suggestions

Executing tasks on the critical path in parallel also allows you to compress the Project Completion timeline. The following table shows some suggested task overlaps for Project Completion.

ID	Primary Task Name	ID	Predecessor Task Name	Overlap
Resource Management				
RM.080	Release Staff	RM.090	Release Physical Resources	75%
RM.090	Release Physical Resources	CR.080	Secure Client Acceptance	100%
Quality Management				
QM.050	Perform Quality Assessment	CR.080	Secure Client Acceptance	75%
Configuration Management				
CM.070	Conclude Configuration Management	CR.080	Secure Client Acceptance	75%

Project Completion task durations can be very dependent on contractual arrangements and the progress of project acceptance. For example, the amount of effort to transfer the CM environment to the client is highly project-specific.

Integrate these tasks with those of other tasks, and examine the time required to conduct them carefully. If the period of rework to complete project deliverables will be uncertain or unusually long, then consider separate consulting tasks, and even a separate agreement, to maintain necessary project support during this time.

APPENDIX

A

PJM Roles

This appendix defines roles used in PJM.

Role Descriptions

Bid Manager

The bid manager is the role responsible for preparation of the bid, negotiation, and award. This role assists in the hand over of materials and information accumulated during the bid to the project manager at the start of the project.

Client Project Manager

The client project manager is responsible for the daily management of the client's contractual commitments to the project. This role must understand the client's business objectives for the project so that these can form the basis for resolving problems, conflicts of interest, and making compromises.

The client project manager obtains physical resources such as accommodation space, office equipment, computer equipment, materials and also arranges for client staff to make time available to the project. The client project manager introduces the consulting staff to other client staff. This role is responsible for monitoring the project's performance: progress against milestones, appropriateness of work, quality of work, and seeks to resolve any problems with work or relationships between development and business staff. The client project manager usually performs intermediate and phase-end acceptance.

Consulting Business Manager

The consulting business manager is the practice manager in the consulting organization who is responsible for the successful execution of the project. This role manages the practice which makes consulting staff available, and coordinates resources needed by the project with other practices. The consulting business manager represents the consulting organization in the contractual agreement with the client, and resolves business issues with the project sponsor. This role participates in project reviews with the client as well as internal practice reviews of project progress.

Project Manager

The project manager is ultimately held responsible for the success or failure of the project. This role must understand the project business objectives of both the client and consulting, and have a clear vision of how to achieve those objectives. The project manager agrees on the scope of the project with the client and resolves any conflicts among the various objectives of its stakeholders.

The project manager is responsible for planning the project, resourcing that plan, and monitoring and reporting the project's progress against the plan. This role obtains any physical resources required for the project, recruits staff, and, if necessary, dismisses staff. The project manager is responsible for ensuring that quality activities are performed in accordance with the Project Management Plan.

Internal responsibilities of the project manager role should be delegated to subordinate team leaders, as documented in the project organization plan.

Project Sponsor

The project sponsor holds the budget and pays for the project. A role at senior management level usually performs this role, and on large, cross-functional projects may be at the board level. This role must have clear views of the objectives of the project, particularly concerning delivery of the business benefits. The project sponsor is the ultimate arbiter on conflicting business requirements, if these cannot be resolved at lower management level, and on scope changes. The project sponsor expects the project to be delivered on time and within budget.

The project sponsor is responsible for ensuring other members of the management share his or her commitment to the project. This role may provide the resources, particularly staff time, required to make the project a success. The project sponsor usually performs the final approval.

Project Support Specialist

The project support specialist is responsible for assisting the project manager in the daily management of the project. On larger projects, this role may be performed by project office staff.

Specific duties are delegated by the project manager but will normally include some or all of the following:

- establish and maintain the Quality Plan, project standards, and project procedures
- coordinate and execute effective phase management
- establish and maintain the Workplan and Finance Plan
- procure staff or physical resources for the project
- monitor and perform analysis of risks, issues, and problems for trends requiring project manager corrective action
- perform coordination and communication functions within the project organization
- organize the Project Library, assign documents into the library, and maintain control of documents in the library
- orient new project members to the project environment, policies and procedures
- coordinate with administrators in client and subcontractor organizations
- prepare project progress reports
- record and distribute minutes, decisions and actions from management meetings
- generate routine status information from project records
- maintain information on project staff such as grade, qualifications, training, parent business unit or subcontractor, telephone and address, project assignment history, and other pertinent information
- develop, document, and implement Configuration Management plans and procedures
- establish project baselines and determine the content of project releases
- ensure that no unauthorized changes are made to a project baseline

- enforce Configuration Management procedures across all project processes
- establish the Configuration Management Repository and assist in the maintenance and protection against damage or loss
- ensure that the standards and procedures which have been defined for the project in the Project Management Plan are implemented
- ensure that quality reviews and quality audits are conducted as required

Quality Auditor

The quality auditor is responsible for conducting quality audits of the project to include a review of the Project Management Plan. This role should be filled by a role independent of the project staff in the consulting organization. The quality auditor needs training in the audit process. This role prepares for, conducts, and reports on the quality audit or audits undertaken, following up any actions raised.

Reviewer

The reviewer plans, conducts, and reports on the results of a quality review. The reviewer also assists in the resolution of review comments and monitors compliance actions by project staff after the review.

APPENDIX

B

PJM Role Usages

This appendix provides an alphabetical index of the roles used in PJM. The following table lists the roles and the PJM tasks which call for the participation of that role.

PJM Role Usages

* Time not included in estimate.

Role Name	ID	Task Name	%
Bid Manager	CR.010	Establish Scope, Objectives, and Approach	*
Client Project Manager	CR.010	Establish Scope, Objectives, and Approach	*
	CR.020	Define Control and Reporting Strategies, Standards, and Procedures	*
	CR.030	Establish Management Plans	*
	CR.040	Issue/Risk Management	*
	CR.050	Problem Management	*
	CR.060	Change Control	*
	CR.070	Status Monitoring and Reporting	*
	CR.080	Secure Client Acceptance	*
	WM.010	Define Work Management Strategies, Standards, and Procedures	*
	WM.020	Establish Workplan	*
	WM.030	Establish Finance Plan	*
	WM.040	Workplan Control	*
	WM.050	Financial Control	*
	RM.010	Define Resource Management Strategies, Standards, and Procedures	*
	RM.020	Establish Staffing and Organization Plan	*
	RM.025	Create Project Orientation Guide	*
	RM.030	Implement Organization	*
	RM.040	Establish Physical Resource Plan	*
	RM.050	Establish Infrastructure	*
	RM.070	Physical Resource Control	*
	RM.080	Release Staff	*
	QM.010	Define Quality Management Strategies, Standards, and Procedures	*
	QM.045	Support Healthcheck	*
	CM.010	Define Configuration Management Strategies, Standards, and Procedures	*
	CM.020	Document Control	*
	CM.030	Configuration Control	*
	CM.035	Knowledge Management	*
	CM.040	Release Management	*
	CM.050	Configuration Status Accounting	*
	CM.070	Conclude Configuration Management	*
Consulting Business Manager	CR.010	Establish Scope, Objectives, and Approach	*
	CR.030	Establish Management Plans	*
	CR.040	Issue/Risk Management	*
	CR.060	Change Control	*
	CR.070	Status Monitoring and Reporting	*
	WM.030	Establish Finance Plan	*
	RM.020	Establish Staffing and Organization Plan	*
	RM.025	Create Project Orientation Guide	*

Role Name	ID	Task Name	%
Project Manager	CR.010	Establish Scope, Objectives, and Approach	70
	CR.020	Define Control and Reporting Strategies, Standards, and Procedures	30
	CR.030	Establish Management Plans	75
	CR.040	Issue/Risk Management	60
	CR.050	Problem Management	50
	CR.060	Change Control	40
	CR.070	Status Monitoring and Reporting	50
	CR.080	Secure Client Acceptance	75
	WM.010	Define Work Management Strategies, Standards, and Procedures	10
	WM.020	Establish Workplan	10
	WM.030	Establish Finance Plan	10
	WM.040	Workplan Control	50
	WM.050	Financial Control	30
	RM.010	Define Resource Management Strategies, Standards, and Procedures	30
	RM.020	Establish Staffing and Organization Plan	60
	RM.025	Create Project Orientation Guide	40
	RM.030	Implement Organization	60
	RM.040	Establish Physical Resource Plan	20
	RM.050	Establish Infrastructure	20
	RM.060	Staff Control	70
	RM.070	Physical Resource Control	20
	RM.080	Release Staff	80
	RM.090	Release Physical Resources	10
	QM.010	Define Quality Management Strategies, Standards, and Procedures	30
	QM.020	Quality Review	10
	QM.030	Quality Audit	20
	QM.040	Quality Measurement	40
	QM.045	Support Healthcheck	40
	QM.050	Perform Quality Assessment	05
	CM.010	Define Configuration Management Strategies, Standards, and Procedures	20
	CM.020	Document Control	10
	CM.030	Configuration Control	20
	CM.035	Knowledge Management	20
	CM.040	Release Management	10
	CM.050	Configuration Status Accounting	20
	CM.060	Audit Key Deliverables	15
	CM.070	Conclude Configuration Management	20
Project Sponsor	CR.010	Establish Scope, Objectives, and Approach	*
	CR.030	Establish Management Plans	*
	CR.060	Change Control	*
	CR.070	Status Monitoring and Reporting	*
	CR.080	Secure Client Acceptance	*
Project Support Specialist	CR.010	Establish Scope, Objectives, and Approach	30
	CR.020	Define Control and Reporting Strategies, Standards, and Procedures	70
	CR.030	Establish Management Plans	25
	CR.040	Issue/Risk Management	40
	CR.050	Problem Management	50
	CR.060	Change Control	60

Role Name	ID	Task Name	%
	CR.070	Status Monitoring and Reporting	50
	CR.080	Secure Client Acceptance	25
	WM.010	Define Work Management Strategies, Standards, and Procedures	90
	WM.020	Establish Workplan	90
	WM.030	Establish Finance Plan	90
	WM.040	Workplan Control	50
	WM.050	Financial Control	70
	RM.010	Define Resource Management Strategies, Standards, and Procedures	70
	RM.020	Establish Staffing and Organization Plan	40
	RM.025	Create Project Orientation Guide	60
	RM.030	Implement Organization	40
	RM.040	Establish Physical Resource Plan	80
	RM.050	Establish Infrastructure	80
	RM.060	Staff Control	30
	RM.070	Physical Resource Control	80
	RM.080	Release Staff	20
	RM.090	Release Physical Resources	90
	QM.010	Define Quality Management Strategies, Standards, and Procedures	70
	QM.020	Quality Review	60
	QM.030	Quality Audit	20
	QM.040	Quality Measurement	60
	QM.045	Support Healthcheck	60
	QM.050	Perform Quality Assessment	95
	CM.010	Define Configuration Management Strategies, Standards, and Procedures	80
	CM.020	Document Control	90
	CM.030	Configuration Control	80
	CM.035	Knowledge Management	80
	CM.040	Release Management	90
	CM.050	Configuration Status Accounting	80
	CM.060	Audit Key Deliverables	85
	CM.070	Conclude Configuration Management	80
Quality Auditor	QM.030	Quality Audit	60
Reviewer	QM.020	Quality Review	30

APPENDIX

C

References and Publications

This appendix provides a listing of references and publications.

References and Publications

Reference: Project Management Institute, *A Guide to the Project Management Body of Knowledge.* PMI Communications, 1996. ISBN 1-880410-12-5.

Reference: International Organization for Standardization, *ISO Standards Compendium, ISO 9000 Quality Management.* Fifth Edition. ISO, 1994. ISBN 92-67-10206-0.

Reference: Carnegie Mellon University / Software Engineering Institute. *CMU/SEI-93-TR-24, Capability Maturity Model for Software, Version 1.1.* Research Access, 1993.

Reference: Oracle Method, *Custom Development Method Handbook*, Version 2.0, November, 1996. Oracle Part Number C10983.

Reference: Oracle Method, *Application Implementation Method Consulting Handbook*, Version 2.0, December, 1996. Oracle Part Number C11180.

Glossary

A

ABT see APPLIED BUSINESS TECHNOLOGY.

Acceptance The approval, typically by a client or user, of a project deliverable.

Access Control The ability to manage which users or groups of users may have the privilege to retrieve, create, update, or delete data held in a repository, such as a relational database.

Activity A set of tasks related either by topic, dependencies, data, common skills, or deliverables. The next level of organization below a phase.

Actuals Information gathered during a project concerning the actual amount of time, finances or resources expended on a task.

Applied Business Technology (ABT) A company that manufactures tools to profile, estimate, and plan projects. Oracle Services has a global licensing agreement with ABT. Oracle Method makes use of Project Workbench, Methods Architect, and Project Bridge Modeler as its worldwide methods and project management software standard; see also PROJECT BRIDGE MODELER and PROJECT WORKBENCH.

Approach A variation or subset of a method, packaged in order to efficiently support the delivery of a service; see also TECHNIQUE.

B

Baseline 1. A starting point or condition against which future changes are measured. 2. A named set of object versions which fixes a configuration at a particular point in time. A baseline normally represents a milestone or key deliverable of a project; see also CURRENT BUSINESS BASELINE.

Billable Project Expenses The project expenses that are billable to a client; see also PROJECT EXPENSES.

Billable Utilization The utilization that is billable to a client; see also UTILIZATION.

Bottom-Up Estimate A task-level estimate derived by calculating the estimating factors critical to completion of each task; see also ESTIMATING FACTOR.

Budget A plan for determining in advance the expenditure of time, money, etc.

C

Change A deviation from a currently established baseline.

Change Control see ISSUE RESOLUTION MANAGEMENT.

Change Request 1. A request for a change to the required behavior of a system, usually from a user as a result of reviewing current behavior. 2. The mechanism by which a change is requested, investigated, resolved and approved; see also IMPACT ANALYSIS.

Completion Criteria Standards or rules which determine completion of a task to an acceptable level of quality.

Configuration A named set of configuration items. Configurations are used to hierarchically organize configuration items in order to facilitate their management; see also CONFIGURATION ITEM.

Configuration Change The implementation of one or more change requests which leaves the configuration in an internally consistent state; see also IMPACT ANALYSIS.

Configuration Item A deliverable or deliverable component which is placed under configuration management.

Configuration Management The process of managing hardware, software, data, and any other documentation needed during the development, testing, and implementation of information systems.

Consulting Grade Level A grade level assigned to Oracle Services consulting resources used to calculate the cost of a resource's labor; 1-Administrative Assistant to 10-Regional Vice President.

Contingency Work effort allotted in a workplan for all unforeseen but possible occurrences of additional work.

Contribution Margin see MARGIN AMOUNT and MARGIN PERCENTAGE.

Controlled Document A document which constitutes or represents a project deliverable for approval internally or by the client, and is subject to change control.

Cost The amount allotted or spent to acquire, deliver or produce anything, for example, the cost of labor to deliver consulting services, the amount spent on incidental costs to deliver consulting services, the amount spent on hardware, software, etc.

CPM see CRITICAL PATH METHOD NETWORK.

Critical Path Method (CPM) Network A network diagram that shows tasks, their dependency links, and their critical path.

CSI see CPU SUPPORT IDENTIFICATION.

D

Deliverable Something a project must produce in order to meet its objectives. A deliverable must be tangible and measurable.

Deliverable Component A part or section of a deliverable. A deliverable component may be the output of a task step.

Deliverable Guideline A detailed description of a deliverable that includes: detailed description, usage, audience, and distribution, format guidelines, control, template, and samples; see also DELIVERABLE.

Deliverable Template A tool designed to aid in production of a deliverable; a template that gives the format and structure of a deliverable; see also DELIVERABLE.

Dependency 1. An indication that one task cannot begin until another task has ended, or progressed to a certain specified level of completion; see also PREDECESSOR and SUCCESSOR. 2. A relationship between two modeling elements, in which a change to one modeling element (the independent element) will affect the other modeling element. (UML 1.1 Semantics)

E

EF see ESTIMATING FACTOR.

Effort The amount of work, measured in person-hours, to perform a task.

Estimate A preliminary calculation of the time and cost of work to be undertaken. The *construct* option calculates estimates using bottom-up, percent adjustment, or top-down techniques in Project Bridge Modeler; see also BOTTOM-UP ESTIMATE, PERCENT ADJUSTMENT ESTIMATE, WORK ESTIMATE, and TOP-DOWN ESTIMATE.

Estimated Function Point Count 7 Function Points per System Entity + 5 Function Points per System Function.

Estimating Factor (EF) A metric that describes an important project characteristic, used to estimate either the amount of effort that project tasks will take, or project risk or complexity. The best EFs are those that represent counts (number of users, objects); see also BOTTOM-UP ESTIMATE.

Estimating Formula A formula that uses estimating factors to derive an estimate for a task.

Estimating Guideline Text which describes in detail how a task is estimated.

Estimating Model The combination of estimating factors and estimating formulas necessary to completely estimate a route.

Expense The amount of money allotted or spent to cover incidental costs (e.g. travel and living) or the cost of something (hardware, software, etc.) to deliver consulting services.

Expense Reimbursement Expenses for reimbursement by the client either allotted or received.

F

Fee A charge, compensation, or payment for a service or product.

FPE see FUNCTION POINT ESTIMATE.

FTP see FILE TRANSFER PROTOCOL.

Function Point Analysis (FPA) A technique used to estimate system size. Using Function Point Analysis, you can calculate system size based on the number of functional element types in the system you are building, as adjusted by its general system characteristics and your technology productivity factor; see also FUNCTIONAL ELEMENT TYPE.

Function Point Estimate (FPE) An estimate of work effort produced using Function Point Analysis or an equivalent method.

G

Gantt Chart A scheduling tool used to display the status of a project's tasks. A Gantt chart shows each task's duration as a horizontal line. The ends of the lines correspond to the task's start and end dates.

Guideline Text that provides instructions and advice for performing a task and suggests possible approaches.

H

Healthcheck Review The project deliverable with components completed by an external and independent consultant to assess the project's progress with respect to plan and objectives.

I

Impact Analysis The process of understanding the complete effect of a particular change; see also CHANGE REQUEST.

Information Systems (IS) A system for managing and processing information, usually computer-based. Also, a functional group within a business that manages the development and operations of the business' information systems.

IS see INFORMATION SYSTEMS.

Issue A situation or concern which requires a resolution. Some issues, if not addressed, could adversely impact the success of a project.

Issue Resolution Management An Oracle Method sub-process and associated tools providing an efficient and effective means of documenting and resolving scope control changes and issues during the life of a large project or program. This sub-process does not support software configuration management.

Iterated Task A task that is repeated once for each iteration in order to increase the quality of the deliverable to a desired level or to add more detail to the deliverable. Iterated tasks are shown as discrete in the workplan; see also TASK, ITERATION, and ONGOING TASK.

Iteration Indicates the number of times or degree to which a task or task group should be repeated, in order to either increase the quality of the task/group deliverables to a desired level, to add sufficient level of detail, or to refine and expand them on the basis of user feedback. A task or group may be "singly iterated" (the expectation is that it will be performed without repetition in the project), or "multiply iterated" (it will be performed successively, multiple times), resulting in a single deliverable for each task (note this is not true for multiply instantiated tasks). For multiple iterations, it is important for planning purposes to give notes on the number of times or the degree to which a task or group should be iterated, based on experience; see also ITERATIVE BUILD, ITERATIVE DEVELOPMENT, and INCREMENT.

Iterative Build Iterative construction of an information system by means of a cycle of code (or generate), test, review, starting from a prioritized (MoSCoW) list of requirements and guided by user feedback; see also MOSCOW LIST.

Iterative Development The application of a repeating cycle of the same or similar activities performed on the same piece of functionality that improves or grows into completion through the iterations; see also INCREMENTAL DEVELOPMENT.

J

K

Key Deliverable A major deliverable that is usually reviewed with the client, signed off (but not necessarily), and placed under change control; see also DELIVERABLE.

Key Resource A person with a wide range of skills or experiences who can be effective in many types of tasks, or is critical to the completion of a specific task.

L

Labor Cost see COST.

Labor Cost Rate The rate (internal cost) for each consulting grade level for delivering services to a client.

Labor Fee see FEE.

Labor Fee Rate The rate (price) for each consulting grade level charged for delivering services to a client.

M

Management The process of planning, controlling, and completing the execution of an undertaking.

Margin Amount The difference between the costs (labor, expenses, and overhead) and revenue plus expense reimbursements expressed as a currency amount.

Margin Percentage The ratio between the margin amount and costs, expressed as a percentage.

Method 1. A system of doing things and handling ideas. A method establishes a network of common tasks, a vocabulary, a set of common processes, estimates, and guidelines for delivering services. 2. The implementation of an operation. It specifies the algorithm or procedure that effects the results of an operation (UML 1.1 Semantics).

Multiply Instantiated Task A task that may be performed at various times during a project, such as status reports or healthchecks, or a task that is partitioned on a project plan, for example, by multiple teams or functional areas.

Multiply Occurring Task A task that is repeated at specific, known times during a project. A multiplying occurring task is similar to a multiply interated task, except the specific occurrences of the task can be planned at the method level. Each occurrence has specific task dependencies which determine when it occurs.

N

Non-Billable Project Expenses The project expenses that are not billable to a client; see also PROJECT EXPENSES.

Non-Billable Utilization The utilization that is not billable to a client; see also UTILIZATION.

O

OM see ORACLE METHOD.

Ongoing Task A task that occurs continuously throughout a project, rather than at a specific known time. Ongoing task are shown as continuous on the workplan; see also TASK, PHASE and ITERATED TASK.

Oracle Method (OM) Oracle Services' integrated service methodology which consists of workplans, handbooks, and templates used to provide enterprise business system solutions.

Oracle Services An Oracle Corporation business organization that provides professional services.

Overhead The operating expenses associated with delivering services, such as rent, light, heat, taxes and non-billable utilization.

Overhead Factor A rate multiplier associated with a category of overhead, e.g., Corporate, Division, or Practice; see also OVERHEAD.

P

Payment Milestone A significant project event at which time a payment is due. Payment milestones can be progress points, dates, the completion of a task or the production of a deliverable.

Payment Terms The terms and conditions upon which payments will be received from a client.

PBM see PROJECT BRIDGE MODELER.

Phase A chronological grouping of tasks in an approach. Services are delivered by phase in order to reduce project risk. Each phase allows a checkpoint against project goals, and measurement against quality criteria to be made.

Phase Control The project management tasks which execute concurrently with phase execution, and perform project monitoring, directing, and reporting functions during a phase.

Phase Execution The method execution tasks performed during a project phase.

Phase Management The project management tasks required to plan, control and complete the execution of a project phase.

Phase Planning The project management tasks which update project plans and procedures for a phase and secure additional resources necessary to execute that phase.

Plan A scheme, method or design for the attainment of some objective or to achieve something.

Predecessor A task that precedes another task and is related to it by a task dependency; see also SUCCESSOR.

Prerequisite Something needed by a task, which is produced by a previous task or an external source; see also DELIVERABLE.

Problem A perceived variance between the expected and observed ability of an item to fulfill its defined purpose.

Problem Report The mechanism by which a problem is recorded, investigated, resolved, and verified.

Process 1. The sequential execution of functions triggered by one or more events. 2. A discipline or sub-project that defines a set of tasks related by subject matter, required skills and common dependencies. A process usually spans several phases in an approach. Examples are: Data Conversion, Testing, Documentation; see also BUSINESS PROCESS and SYSTEM PROCESS.

Program 1. A set of coded instructions that a computer executes or interprets to perform an automated task. 2. A interrelated group of projects that are either being run concurrently or sequentially and that share a system goal. Individual projects may have different goals, however the combined set of projects will have a program goal.

Project Bridge Modeler (PBM) Applied Business Technology tool used to build the project planning and estimating system. PBM allows project managers to select, edit, combine, and create project workplans or routes, that best fit the needs of the client.

Project Completion The third and final part of the PJM project life-cycle. The satisfactory conclusion of the project and settlement of all outstanding issues prior to hand over of the project deliverables to the client.

Project Earned Value A measure of the value of completed tasks in a project. There are various ways of measuring the value of a task. These include percentage on commencement, percentage on completion, and amount at milestone.

Project Expenses Funds allocated or spent to cover incidental or non-labor costs of a project.

Project Infrastructure The framework for storing, maintaining, and referencing all implementation deliverables and supporting materials including office space, software tools, and standards.

Project Library 1. A system for storing, organizing and controlling all documentation produced or used by the project. 2. The physical location of all deliverables for a single project, plus administrative and support materials. An administrative office to which all members of a team have access.

Project Life-Cycle The organization of a project according to its three major parts: planning, execution, and completion.

Project Management Plan Initially, a definition of the project's scope, objectives, and approach. The management approach is subsequently refined during project and phase planning with additional detail and supported by necessary standards and procedure.

Project Milestone A significant project event.

Project Objectives The set of criteria for measuring a project's success.

Project Office The management of the project library for a specific project.

Project Orientation Guide A key project deliverable which establishes the resource policies and procedures for items such as travel and living expenses.

Project Planning The first part of the PJM project life-cycle. The definition of a project with respect to scope, quality, time and cost. Project planning also determines the appropriate organization of resources and responsibilities to execute a project.

Project Schedule A list of tasks to be carried out presented against a timetable for their completion.

Project Timeline A specification of work to be carried out together with the number of resources needed to achieve a target duration; see also PROJECT SCHEDULE.

Project Workbench (PMW) Applied Business Technology tool used to schedule, track, and analyze your project. PMW provides work breakdown structures based on the routes selected in Project Bridge Modeler and provides the capability to manage these plans.

Project Workplan A specification of the work to be performed for a project, expressed as a set of interdependent tasks with project resources allocated over time.

Promotion 1. The state change of an item within its life-cycle representing the successful completion of a quality requirement such as a test, review or management authorization. Promotion reflects increasing value and confidence in the item, indicating a need for stricter control of changes to the item.
2. The raising of a configuration item to a higher promotion level; see also DEMOTION.

PMW see PROJECT WORKBENCH.

Q

Quality Audit An audit used to assess the adherence of the project team to plans, procedures, and standards.

Quality Review A review used to assess the quality of a deliverable in terms of fitness for purpose and adherence to defined standards and conventions.

R

Release A baseline issued from the CM Repository for delivery to an a destination. The destination may be internal to the project environment, such as for testing, or external, such as to the client.

Repository A mechanism for storing any information about the definition of a system at any point in its life-cycle. Repository services would typically be provided for extensibility, recovery, integrity, naming standards, and a wide variety of other management functions.

Resource Any persons, equipment, or material needed to perform a task(s).

Resource Category see CONSULTING GRADE LEVEL.

Revenue Income from services labor fees, training fees, licensed product sales and support fees.

Revision The authorized modification to a configuration item.

Risk 1. The potential of an adverse condition occurring on a project which will cause the project to not meet expectations. A risk requires management assessment and a strategy for its mitigation. 2. The logical product of the impact of the risk and the likelihood of it occurring.

Role A classification of staff member used on a project. Examples are: analyst, application developer, system architect; see also RESOURCE.

Role Placeholder A fictitious resource name with a resource category assigned to replace a role; e.g., Analyst 1 - Principal, Analyst 2 - Senior.

Route A variation of a method containing all tasks required in order to deliver a service; a dependency network.

S

Scope The boundaries of a project expressed in some combination of geography, organization, applications or business functions.

Scope Change A change to project scope. A scope change requires an adjustment to the project workplan, and nearly always impacts project cost, schedule or quality.

Scope Creep The common phenomenon where additional requirements are added after a project has started without reconsidering the resourcing or timescale of the project. Scope creep arises from the misapprehension that such small additions will not affect the project schedule.

Sign-off Agreement with a client of the successful completion of a project, project phase, or deliverable.

Singly Instantiated Task A task which occurs once, at a specific time, during a project.

Site A uniquely identifiable geographic location or place from which one or more business organizations may be wholly or partly operating.

Stakeholder A person, group, or business unit that has a share or an interest in a particular activity or set of activities.

Standard A set of rules for measuring quality. Usually, standards are defined for products deliverables or deliverable components and processes.

State A recognizable or definable condition that a system or an object can be in at some point in its life-cycle.

State Transition A valid change of a system or an object from one state to another, modeled on a state transition diagram; see also STATE.

Sub-process A process performed entirely within another process.

Successor A task that follows another task and is related to it by a dependency link; see also PREDECESSOR.

T

Task A unit of work that done in delivering a service. A task is the smallest trackable item on a project plan, and forms the basis for a work breakdown structure. The minimum elapsed time for a task iteration or instantiation should be one day. The maximum should be two weeks; see also WORK BREAKDOWN STRUCTURE.

Task Dependency The relationship between two tasks where the start or end date of the successor task is constrained by the start or end date of the predecessor task.

Task Dependency Network A network of tasks and task dependencies where each node is a task and each link is a task dependency; see also CRITICAL PATH METHOD (CPM) NETWORK.

Task Step A discreet step to be done in executing a task.

Technique A specific approach to performing a task. A methodical means of handling and communicating complex details.

Timebox A project management technique that a fixes the duration and resources of a task, or set of tasks, and forces the scope of the project to be adjusted based on the time available to complete the task(s). The contingency for under-estimation of the work is provided by a prioritized list of features left out if necessary. The contingency for over-estimation is provided by a prioritized list of features that should be added in if time allows.

Tool Software applications, deliverable templates, or any other utility suggested to facilitate the completion of a particular task.

Top-Down Estimate A high-level work effort estimate. This type of estimate is derived by taking a total project estimate and dividing it among the project's phases, activities, and tasks.

U

Uncontrolled Document A document which is produced once for information only, and is not subject to formal approval or change control.

User A person who uses a system to perform a business function.

Utilization The amount of time a staff member books to a project accounting system; see also BILLABLE UTILIZATION and NON-BILLABLE UTILIZATION.

V

Variance The difference between a planned and an actual value; for example, budgeted hours vs. actual hours.

Version The rendering of a configuration item which incorporates all of its revisions starting from a given point.

Version Control A mechanism to manage multiple revisions of files, documents, programs, applications, or other items that undergo change.

View A means of accessing a subset of data in a database.

W

WBS see WORK BREAKDOWN STRUCTURE.

Work Breakdown Structure (WBS) An organization of project tasks into a hierarchy for scheduling and reporting progress.

Work Effort see EFFORT.

Work Estimate see ESTIMATE.

Work Schedule see PROJECT SCHEDULE.

X

Y

Z